# Pre-Stall Load Redistribution on Swept Tailplane Models with Full-Span Controls

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**Local loading changes on a swept lifting surface with a deflected plain control can alter lift growth and pitching moment well before maximum lift. MONNALISA wind-tunnel measurements and Reynolds-averaged Navier–Stokes calculations are used to examine how fixed-surface and full-span control loads combine in this response. For the positive-sweep, zero-dihedral model at $\delta = 20°$, the computed fixed surface continues gaining lift while both surface contributions fall below their preceding incidence trends. Across the tested baseline windows, the fixed surface accounts for about 63% of the total lift departure; the control contributes more strongly to the moment departure about the fitted aerodynamic center. Pressure sections and regional integrations connect these different allocations to stronger leading-edge suction, reduced hinge/aft loading, and larger outboard moment weights. Matched dihedral comparisons distinguish absolute load projection from changes with incidence. The forward-swept planform gives weaker aft losses and a different spanwise moment allocation. The detailed interpretation concerns the computed states, for which correspondence with measured integrated coefficients is partial. The combined force and moment analysis identifies loading changes that continuing total lift growth alone would conceal.**

## Nomenclature

| | | |
|---|---|---|
| $C_D, C_L$ | = | drag and lift coefficients |
| $C_{F,k}$ | = | body-force coefficient, $k = x, y, z$ |
| $C_M^A, C_M^O, C_M^{A_I}$ | = | moments about aerodynamic center, origin, or common configuration-I axis |
| $C_p,\ \mathbf{C}_f, C_{f,c}$ | = | pressure, vector skin-friction, and chordwise skin-friction coefficients |
| $D_{Q,j}$ | = | departure from the preceding coefficient trend |
| $F_k$ | = | body-force component, N |

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$F_n, F_s$ = panel-normal and local spanwise force, N

$\mathcal{I}_M$ = dihedral contrast of moment increments

$M_\infty$ = freestream Mach number

$Q$ = coefficient label, $L$ or $M$

$Re_b, Re_{\bar{c}}$ = semispan and mean-chord Reynolds numbers

$S$ = undeformed semiplanform reference area, $\mathrm{m}^2$

$T_\infty$ = freestream temperature, K

$U_\infty$ = freestream speed, m/s

$b$ = undeformed semispan, m

$c,\ \bar{c}$ = local and mean aerodynamic chords, m

$f_{\mathrm{rev}}$ = upper-aft area fraction with reverse chordwise wall shear

$i, j, k$ = sample, surface-contribution, and force-component indices

$m_{Q,j}^{\mathrm{pre}}$ = preceding slope, per degree

$p, p_\infty$ = surface and freestream pressures, Pa

$q_\infty$ = freestream dynamic pressure, Pa

$s$ = undeformed span coordinate, m

$t$ = section thickness, m

$\mathbf{t}_c$ = downstream chordwise unit tangent

$u_i, w_i$ = supplied sample uncertainty magnitude and contrast weight

$u_{\mathrm{ind}}, u_{\mathrm{L1}}$ = independence and absolute-weight contrast sensitivities

$x, y, z$ = downstream, spanwise, and upward body coordinates, m

$y^+$ = nondimensional wall distance

$x_A, x_O$ = aerodynamic-center and origin abscissae, m

$\alpha, \alpha_a, \alpha_b$ = incidence and interval endpoints, deg

$\Gamma, \delta$ = dihedral and control deflection, deg

$\Delta C_{Q,j}$ = ordinary coefficient increment

$\gamma_{\mathrm{BC}}$ = algebraic transition-model intermittency factor

$\eta = s/b,\ \xi$ = span fraction and straightened local chord fraction

$\lambda, \Lambda$ = quarter-chord sweep (deg) and aspect ratio

$\rho_\infty$ = freestream density, kg/m$^3$

$\tau, \boldsymbol{\tau}_w$ = taper ratio and wall-shear vector (Pa)

# I. Introduction

A deflected trailing-edge control changes both the amount of lift generated by a lifting surface and the location at which that load acts. Its response with incidence is therefore relevant to pitching moment as well as lift, even before the maximum-lift condition is reached. When local loading changes, total lift can continue increasing while its growth slows and the pitching response changes more abruptly. Interpreting this behavior requires relating the integrated coefficients to the loads on the fixed and moving parts of the surface.

Sectional plain-flap experiments establish the starting point for this problem. Sears and Liddell [1] measured lift, drag, pitching moment, and hinge moment for a 30%-chord plain flap on an NACA 0015 airfoil, including sealed and unsealed hinge gaps. Separation reduced flap effectiveness at large deflections, although additional deflection still increased lift within the tested range. Their comparison with a thinner symmetric section also showed that similar lift effectiveness could coexist with different hinge-moment behavior. Reduced flap effectiveness is thus established physics; neither the presence of separation nor a single integrated lift trend fully characterizes the loading response.

On a finite wing, flap-induced loading is also shaped by the spanwise flow and planform. Campbell, Blanks, and Leaver [2] investigated rectangular wings of small aspect ratio with full-span plain flaps of several chord fractions. Their force and pitching-moment measurements, compared with thin-wing theory and empirical thickness and boundary-layer allowances, related flap effectiveness and center of pressure to flap chord and aspect ratio. Those unswept, thinner, transition-tripped models differ from the present swept models, but they establish why the location of the additional load matters alongside its magnitude.

For swept wings, Harper and Maki [3] documented how spanwise changes in local lift response can produce pitching-moment nonlinearity below maximum lift. The first appearance of local stall need not coincide with the first integrated moment change, because the force change and its moment arm vary along the span. Their broad stall review provides context for examining load redistribution, rather than an equivalent test of the present deflected-control configurations.

Nonlinear sectional data provide one route for representing these responses in reduced-order models. Sivells and Neely [4] coupled such lift data to lifting-line theory and also calculated wing pitching moment from sectional moment data. Goitia and Llamas [5] coupled a vortex-lattice formulation to viscous sectional data, with effective-sweep and finite-wing corrections, to predict nonlinear wing and tail lift. Consequently, the ability of a reduced-order model to

reproduce the response depends on how sectional nonlinearities and three-dimensional loading changes are represented.

The immediate basis for the present work is the MONNALISA database publication [6]. Its Sec. IV and Fig. 11 already document the baseline pre-stall lift nonlinearity, expansion of reversed flow on the control, associated pressure changes, and continued growth of total lift. Its DUST formulation uses sectional tables and corrections, while the retained potential-based panel and vortex-lattice predictions are smoother. The present loading analysis builds on these established observations.

The remaining loading question is how changes on the fixed surface and control combine to produce the joint lift and moment response. Continued fixed-surface lift growth, for example, does not establish that this contribution follows its preceding trend. Comparing ordinary increments with departures from that trend identifies which components supply the reduced total growth, while spatial pressure integrations show why the component ranking can differ for moment. This study develops that quantitative interpretation from the existing Reynolds-averaged Navier–Stokes (RANS) solutions, first assessing their integrated coefficients against the measured response and its supplied uncertainty. Changes in control deflection and geometry then test how broadly the reference loading pattern persists within the dataset.

The test articles are isolated tailplane-inspired wings with full-span, 30%-chord plain controls; no fuselage, wing wake, or complete-aircraft trim condition is represented. Dihedral is varied on the same planform, while a forward-swept planform provides the contrasting sweep direction. These comparisons define the scope of the analysis without establishing behavior for partial-span controls, other flap chords, or intermediate dihedral angles.

## II. Experimental, Numerical, and Analysis Framework

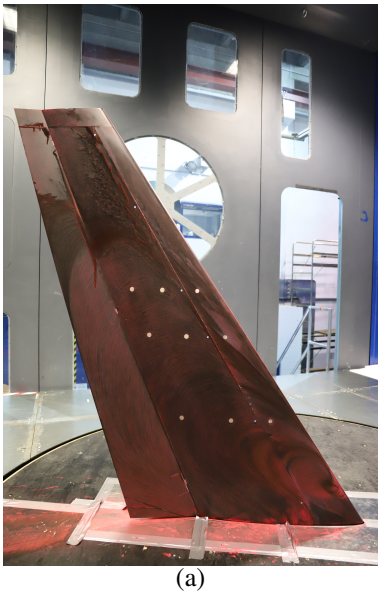
### A. Models and experimental installation

Wind-tunnel (WT) measurements establish the integrated lift and pitching-moment response, while RANS surface loads identify how the fixed surface and control contribute to it. To examine dihedral, the reference aft-swept planform (configuration I) is investigated at $\Gamma = 0°$ and $\Gamma = 45°$, with its in-plane geometry unchanged. A forward-swept planform (configuration II) provides a contrast in sweep direction and overall planform response. The labels I and II are used in the tables.

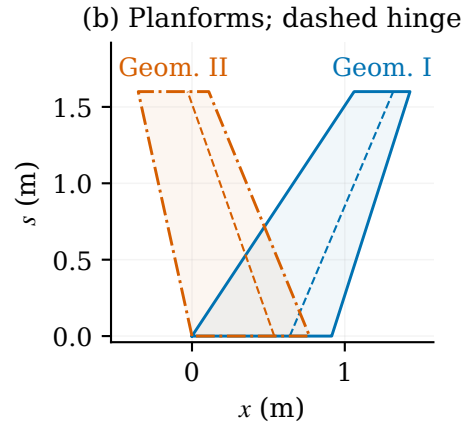
Both models use a symmetric section of constant profile along the undeformed semispan $b = 1.6$ m, with thickness ratio $t/c = 9\%$, where $c$ is the local chord. Each model has a full-span plain control hinged at 70% chord. The reference surface $S$ is the undeformed semiplanform area, and $\bar{c}$ denotes its mean aerodynamic chord (MAC). Table 1 gives the geometric parameters and coefficient references. The quarter-chord sweep is $\lambda = +30°$ for configuration I and $\lambda = -15°$ for configuration II; positive dihedral $\Gamma$ raises the tip. Aspect ratio is defined for the corresponding full planform as $\Lambda = 2b^2/S$, and $\tau$ is the tip-to-root chord ratio. Configuration II also differs in taper, aspect ratio, MAC and chord-based Reynolds number, so the sweep contrast is interpreted through the overall planform response.

**Table 1 Model geometry and coefficient reference quantities.**

| Property | I, $\Gamma = 0°$ | I, $\Gamma = 45°$ | II, $\Gamma = 0°$ |
|---|---|---|---|
| $\lambda$ (deg) | 30 | 30 | −15 |
| $\Lambda$ | 5.0 | 5.0 | 5.2 |
| $\tau$ | 0.4 | 0.4 | 0.6 |
| $S$ (m$^2$) | 1.024 | 1.024 | 0.984604 |
| $\bar{c}$ (m) | 0.679184 | 0.679184 | 0.6282 |
| Root chord (m, rounded) | 0.914286 | 0.914286 | 0.76923 |
| $Re_{\bar{c}}$ | $2.335 \times 10^6$ | $2.335 \times 10^6$ | $2.159 \times 10^6$ |
| Numerical $x_A$ (m) | 0.642000 | 0.643532 | −0.003618 |



(a)



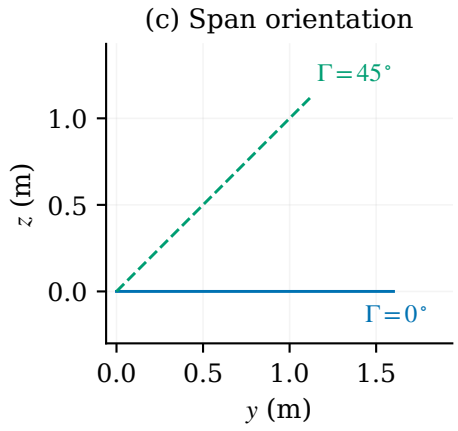




**Fig. 1 Wind-tunnel installation and geometry: (a) mounted semispan model; (b) reference and forward-swept planforms; (c) zero- and positive-dihedral arrangements.**

The measurements were obtained in the low-turbulence section of the Galleria del Vento del Politecnico di Milano (GVPM), a closed-circuit facility. The test section is 4.00 m wide and 6.00 m long, with an original height of 3.84 m. For the present tests, installation of the false floor reduced the test-section height to 3.49 m; the undeformed model semispan is 40% of its width. A curved upstream inlet and gap beneath the false floor were designed to divert the incoming tunnel-floor boundary layer below the model installation. Boundary-layer development over the raised floor and around the support remains distinct from a numerical symmetry boundary. The installation photograph in Fig. 1 was taken at $\alpha = 25°$, $\delta = 20°$ and illustrates the mounting, rather than the pre-stall state.

The database setup description [6] gives freestream turbulence below 0.1%, measured using a turbulence sphere and hot-wire anemometry. Outside the wall layers, the inlet total-pressure variation was within ±0.5% of dynamic pressure over most of the section and within ±1% overall; outlet measurements confirmed similar uniformity. The reported longitudinal static-pressure gradient is $\partial C_P/\partial x = -0.004\ \mathrm{m}^{-1}$. Loads were acquired through a RUAG 767 strain-gauge balance below the floor and an HBM QuantumX system; the reduction includes the balance-to-wind transformation and the database corrections. The nominal freestream Mach number and temperature are $M_\infty = 0.147$ and $T_\infty = 300$ K, with MAC Reynolds numbers given in Table 1. The common database operating point is $Re_b = 5.5 \times 10^6$; the MAC-based values are $Re_{\bar{c}} = Re_b \bar{c}/b$.

Recorded incidences are $\alpha = -25, -24, -22, \ldots, 22, 24, 25°$: the local pre-stall spacing is $\Delta\alpha = 2°$, with $\Delta\alpha = 1°$ endpoint steps. The comparisons use $\delta = 0, 10, 20, 30°$. Uncertainty bars are the supplied propagated measurement/reduction-chain magnitudes, including balance and transformation contributions. Each moment about the corresponding aerodynamic center retains its associated propagated uncertainty. The contrast sensitivities below use these supplied magnitudes.

### B. Conventions and moment references

Positive incidence $\alpha$ raises the leading edge relative to the freestream, and positive control deflection $\delta$ moves the trailing edge downward relative to the fixed section. Lift is positive toward the upper surface and pitching moment is positive nose-up. The WT and numerical data are expressed in these common physical conventions. In body axes, $x$ is downstream, $y$ is spanwise, and $z$ is upward. The axial force is the downstream body-axis component $F_x$, and the body-normal force is the upward component $F_z$. The latter is defined in the common body frame, rather than normal to a panel tilted by dihedral. The body-force components $F_k$, with $k = x, y, z$, are normalized by the freestream dynamic pressure $q_\infty = \rho_\infty U_\infty^2/2$ and area $S$, where $\rho_\infty$ and $U_\infty$ are freestream density and speed. Lift and drag coefficients follow from

$$C_L = C_{F,z}\cos\alpha - C_{F,x}\sin\alpha, \qquad C_D = C_{F,z}\sin\alpha + C_{F,x}\cos\alpha, \qquad C_{F,k} = F_k/(q_\infty S). \tag{1}$$

Pitching moment is normalized by $q_\infty S\bar{c}$. Body coordinates are expressed relative to the root leading edge.

Both experimental and numerical moments are referred to aerodynamic centers determined from the zero-deflection loading characteristics. Linear fits to moment and body-normal force over $|\alpha| \leq 10^\circ$ define the positions, which remain fixed when control deflection varies [6]. The experimental fits use the supplied uncertainty weights; the numerical fits use the retained zero-deflection linear trends. For configuration I, the WT and RANS positions are 0.6430 and 0.6420 m at zero dihedral, and 0.6455 and 0.6435 m at $\Gamma = 45^\circ$: separations of about 1.0 and 2.0 mm. For configuration II they are $-0.0492$ and $-0.00362$ m, separated by 45.6 mm. The comparison coefficient $C_M^A$ therefore uses the same reference construction, with each dataset's fitted position; it is not an imposed quarter-MAC moment.

The original numerical reference is $O = (0.228571, 0, 0)$ m, the root quarter-chord of Geom. I. For a numerical pole transported from abscissa $x_0$ to $x_1$,

$$C_M(x_1) = C_M(x_0) + \frac{x_1 - x_0}{\bar{c}} C_{F,z}. \tag{2}$$

The fitted abscissae are in Table 1. Matched dihedral diagnostics use the identical numerical axis $A_I = (0.642000, 0, 0)$ m in both configurations, denoted $C_M^{A_I}$. Geom. II uses its own MAC once for every moment contribution; all coefficients use the fixed areas in Table 1.

### C. Numerical methodology

The SU2 [7] calculations solve compressible ideal-gas RANS with Sutherland viscosity, the Spalart–Allmaras (SA) turbulence model [8], and the Bas–Cakmakcioglu (BCM) algebraic transition treatment [9]. The finite-volume mean-flow discretization uses median-dual vertex cells and Roe fluxes with second-order monotonic upstream-centered schemes for conservation laws (MUSCL) reconstruction [10], Green–Gauss gradients, and a Venkatakrishnan limiter [11] with coefficient 0.05. SA convection uses first-order scalar upwinding. The BCM intermittency factor $\gamma_{\mathrm{BC}}$ modifies turbulence production without an additional transported transition equation.

The model surfaces have no-slip, adiabatic boundaries, with freestream conditions at the far field and symmetry at the root. The symmetry plane omits the physical support and boundary-layer development over the raised-floor arrangement. Steady solutions use implicit Euler pseudo-time advancement and flexible generalized minimal residual (FGMRES) linear solves, with adaptive Courant–Friedrichs–Lewy (CFL) number. The production stopping criterion monitors lift and drag over 100 entries with a Cauchy tolerance of approximately $3 \times 10^{-6}$. Case-specific solution sensitivity is discussed where it affects the interpretation.

The surface grid resolves the leading edge, hinge junction, and trailing edge, where curvature and pressure changes require additional resolution. Wall-normal boundary-layer extrusion uses a nominal first-cell height of about $2 \times 10^{-6}$ m and growth ratio 1.15, while incidence-aware volume refinement follows the wake. The volume mesh is hex-dominant and octree-based [6]. Representative solutions have median nondimensional wall distance $y^+ = 0.19$–$0.26$, with local

maxima about 1.1–1.3. A reference grid contains approximately 9.44 million points and 18.92 million volume elements.

The existing grid-refinement assessment uses four levels—coarse, nominal, fine, and very fine—for configuration I at $\delta = 35°$, $\Gamma = 0, 45°$, and $\alpha = 0, 18°$. Table 2 gives the point and element ranges. Figure 2 compares lift, drag, and pitching moment against the actual grid-point count on a logarithmic scale; open squares identify the nominal level, and lines connect the retained levels. Moments are about the numerical origin $O$, using the configuration-I area and MAC.

The nominal level was selected to balance spatial resolution with the demands of the extensive MONNALISA campaign across geometries, incidences, and control deflections. Its approximately 9.7–12.7 million points compare with 50.1–65.9 million on the finest level. Nominal-to-finest differences are 1.20–2.14% for lift and 1.13–2.49% for drag. At zero dihedral and $\alpha = 18°$, $C_M^O$ changes from approximately −0.813 to −0.827, an absolute difference of 0.0134; the $\Gamma = 45°$ moment sequence is nonmonotonic. The assessment supports the nominal grid as a practical production compromise while showing finite, condition-dependent integrated-load sensitivity. It provides no discretization-error bound for local component loads. The comparison with measurements in Sec. III separately assesses reproduction of the principal pre-stall aerodynamic trends.

**Table 2 Point and volume-element ranges, in millions, for the four-level refinement assessment.**

| Refinement | Points | Volume elements |
|---|---|---|
| Coarse | 4.79–5.29 | 9.51–10.90 |
| Nominal | 9.67–12.74 | 19.06–24.07 |
| Fine | 22.33–26.56 | 39.93–45.70 |
| Very fine | 50.11–65.89 | 87.35–109.98 |

### D. Additive loading and empirical departures

A surface can gain lift while contributing to a slowdown in total lift growth. To distinguish these effects, the actual increment is the measured or computed coefficient change between two incidences. The expected increment is obtained by linearly extrapolating the empirically fitted preceding slope over the same interval. Their difference, actual minus expected, is the departure. For the fixed surface of configuration I at zero dihedral and $\delta = 20°$, the actual increase over $\alpha = 0 \rightarrow 2°$ is +0.0593, while continuation of the preceding trend predicts +0.1285. The departure is −0.0692: the fixed surface still gains lift, but gains 0.0692 less than predicted by that trend. This is a reduction in growth, not an actual lift loss.

The analysis resolves these changes into additive surface loads. Solver surface groups define the fixed surface (upper/lower fixed panels), the control (upper/lower control panels, hinge junction, and blunt trailing edge), and tip. Pressure and viscous contributions are retained on both physical sides with the same global $S$ and $\bar{c}$. Tip loads remain in every total and closure check. All displayed coefficients use their actual magnitude.

For coefficient label $Q = L$ or $M$ and surface contribution $j$ (fixed, control, tip, or total), the ordinary increment

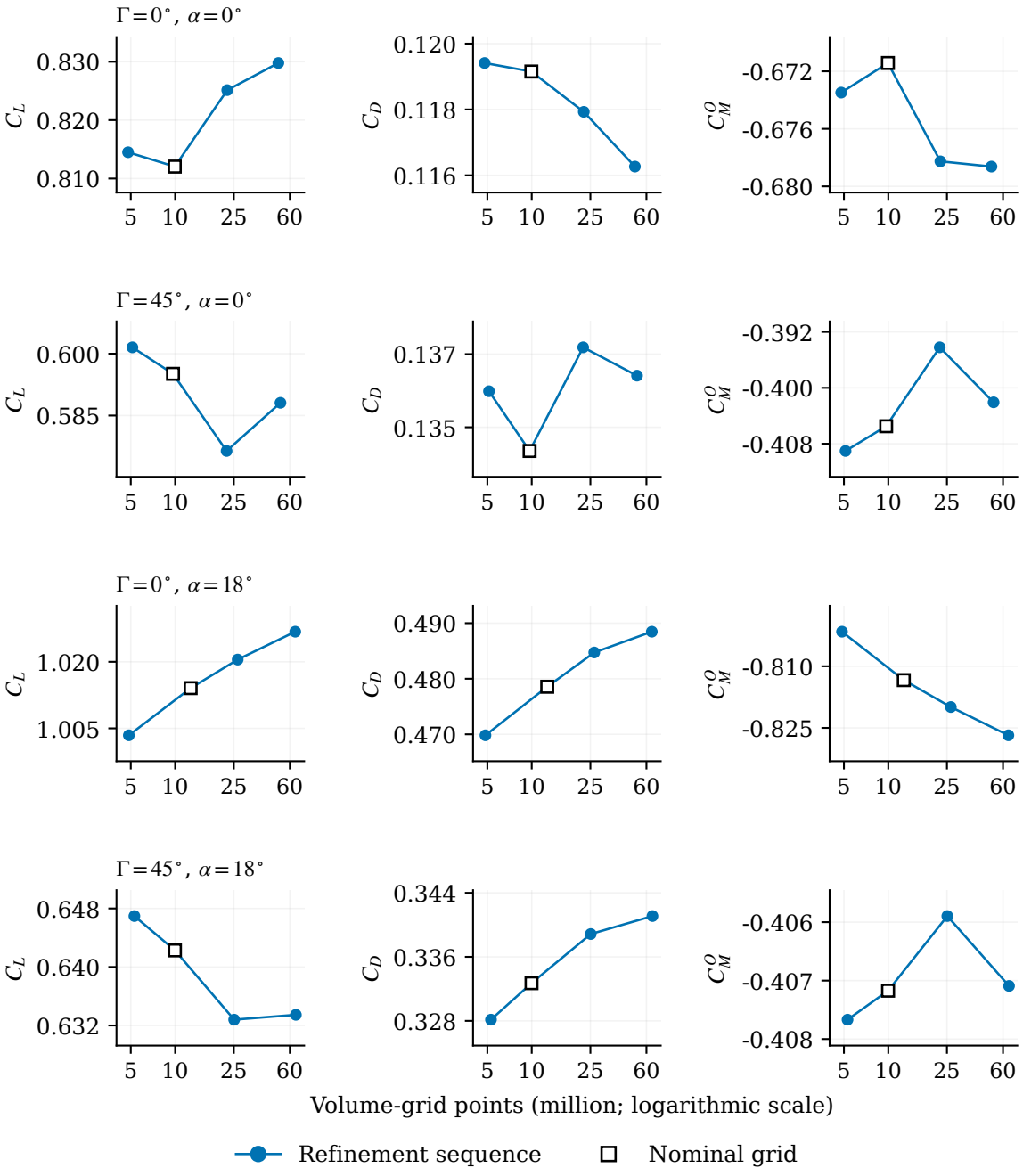



**Fig. 2 Grid-refinement trends of lift, drag, and pitching moment for configuration I at $\delta = 35°$, two incidences, and two dihedral settings.**

is the change between sampled incidences $\alpha_a$ and $\alpha_b$. Its departure subtracts the change predicted by continuing the preceding slope $m^{\mathrm{pre}}_{Q,j}$, measured per degree,

$$\Delta C_{Q,j} = C_{Q,j}(\alpha_b) - C_{Q,j}(\alpha_a), \qquad D_{Q,j} = \Delta C_{Q,j} - m^{\mathrm{pre}}_{Q,j}(\alpha_b - \alpha_a). \tag{3}$$

The preceding slope is an unweighted linear least-squares fit to the declared preceding samples. Using identical samples and weights for every component and the total preserves additivity for both increments and departures. The primary preceding-incidence range for $\delta = 20°$ is $\alpha \in \{-6, -4, -2\}°$; the early reference $\delta = 30°$ numerical baseline is $\alpha \in \{-16, -14, -12\}°$. These are empirical references, with no assumption that the preceding flow is attached. Neighboring preceding-incidence fitting ranges test this reference dependence, with the selected surface solutions held fixed. Where the preceding sequence is curved, signed finite differences describe the response without imposing a unique event amplitude.

The uncertainty sensitivity of each measured lift or moment departure is evaluated by propagating the supplied coefficient uncertainties through its defining linear combination. Equation 3 is written as $D = \sum_i w_i C_i$, where $i$ indexes the measured samples, $w_i$ is the combined coefficient weight, and $u_i$ is its supplied propagated uncertainty magnitude. The independence-based sensitivity $u_{\mathrm{ind}}$ and absolute-weight sensitivity $u_{\mathrm{L1}}$ are

$$u_{\mathrm{ind}} = \left[\sum_i (w_i u_i)^2\right]^{1/2}, \qquad u_{\mathrm{L1}} = \sum_i |w_i| u_i. \tag{4}$$

The first is conditional on independence; the second bounds a perturbation when each sample varies within its supplied magnitude. Coverage and between-incidence covariance are undocumented, so these are contrast sensitivities rather than confidence intervals.

To locate the changes identified by the component budget, pressure and wall-shear forces are integrated over chordwise and spanwise regions. The pressure coefficient is $C_p = (p - p_\infty)/q_\infty$, and the skin-friction vector is $\mathbf{C}_f = \boldsymbol{\tau}_w / q_\infty$, where $p_\infty$ is freestream pressure and $\boldsymbol{\tau}_w$ is wall shear. Linear surface tractions and their moments are integrated over the triangulated surface, with

$$C_M(x_A) = \frac{1}{q_\infty S \bar{c}} \int \left[ z\, dF_x - (x - x_A) dF_z \right]. \tag{5}$$

Here $x_A$ is the moment-reference abscissa. The moment of the local downstream force $dF_x$ is $z\, dF_x$, and that of the local upward force $dF_z$ is $-(x - x_A)dF_z$. These $F_x$ and $F_z$ contributions are retained separately for the common-axis dihedral comparison. Reconstructed full-surface forces and moments agree with the solver-integrated coefficients before decomposition into chordwise and spanwise load contributions; detailed closure checks are retained in the

reproducibility records.

The regions distinguish loading upstream of the hinge from loading on the deflected control, and compare inboard, middle, and outboard contributions. The nondimensional span position is $\eta = s/b$, where $s$ is distance along the undeformed span; $\xi$ is the local chord fraction after straightening the control about its hinge. Chordwise regions are forward ($\xi < 0.4$), intermediate ($0.4 \le \xi < 0.7$), and aft ($\xi \ge 0.7$). Chordwise contributions are summed over the complete semispan, while the aft spanwise contributions use semispan thirds (Fig. 3). Contributions include both physical sides, with end-face loads completing the total. Sectional $C_p$ is shown at $\eta = 0.2, 0.5, 0.8$ on both physical sides. The reference sections use the fixed and control panels, excluding narrow hinge-junction and end faces; the dihedral and forward-swept sections use complete surface intersections, including the junction and blunt trailing edge. Curves follow the connected surface intersections without bridging gaps or smoothing. Regional contributions are integrated loads, rather than load densities.

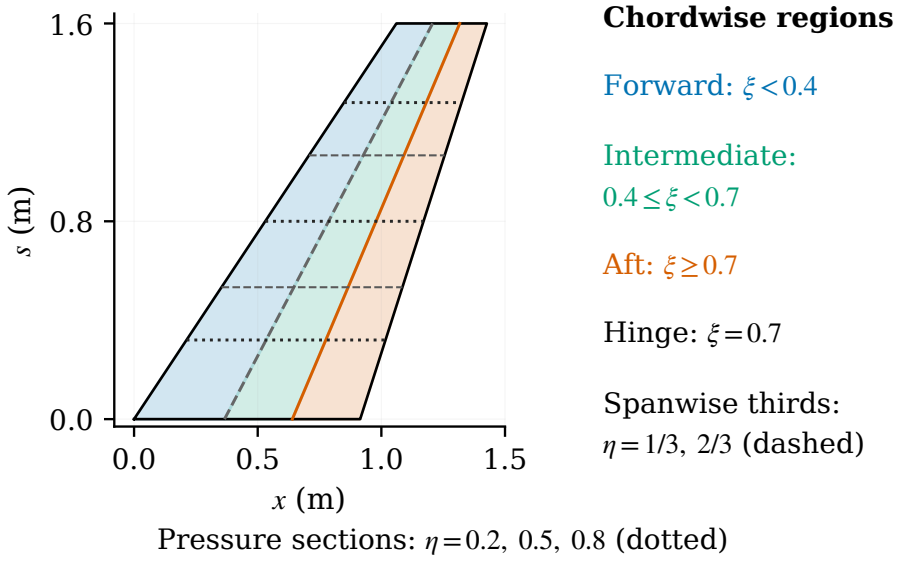



**Fig. 3 Undeformed reference planform: chordwise integration regions, spanwise thirds, hinge, and pressure-section locations.**

The local downstream direction must account for control rotation and surface orientation before wall-shear reversal is assessed. Let $\mathbf{d}_c$ be the downstream chord direction: $(1, 0, 0)$ on the fixed panel, rotated about the physical hinge on the control. On each surface triangle with unit normal $\mathbf{n}$, its tangent-plane projection is normalized to give $\mathbf{t}_c = [\mathbf{d}_c - (\mathbf{d}_c \cdot \mathbf{n})\mathbf{n}]/\|\mathbf{d}_c - (\mathbf{d}_c \cdot \mathbf{n})\mathbf{n}\|$. This construction points downstream on both physical sides, includes dihedral and flap rotation, and is applied after transforming symmetry-reflected states into the common physical convention. End faces are excluded from the reversal measure; the retained upper-control faces have no degenerate projected directions.

The signed coefficient $C_{f,c} = \mathbf{C}_f \cdot \mathbf{t}_c$ is negative where the chordwise component of wall shear points upstream. We report the fraction $f_{\text{rev}}$ of the upper side of the deflected control ($\xi > 0.7$, excluding end faces) satisfying $C_{f,c} < 0$,

weighted by actual surface area. Thus $f_{\mathrm{rev}} = 0.5$ means that upstream-directed chordwise shear occupies half that upper-control area. Linear interpolation within each triangle locates the sign crossing. This area fraction measures reversed near-wall chordwise flow. The additional $\gamma_{\mathrm{BC}}$ field is the algebraic factor multiplying modeled turbulence production; it is used with the shear pattern to interpret the computed boundary-layer development.

## III. Measured Response and Numerical Correspondence

The measurements establish whether lift and pitching moment change together as incidence rises at fixed control deflection. Figure 4 presents the reference zero-dihedral polars. The comparison includes deflections below and above the pronounced lift slowdown so that a change in moment response need not be assigned to the same incidence or deflection as a lift feature.

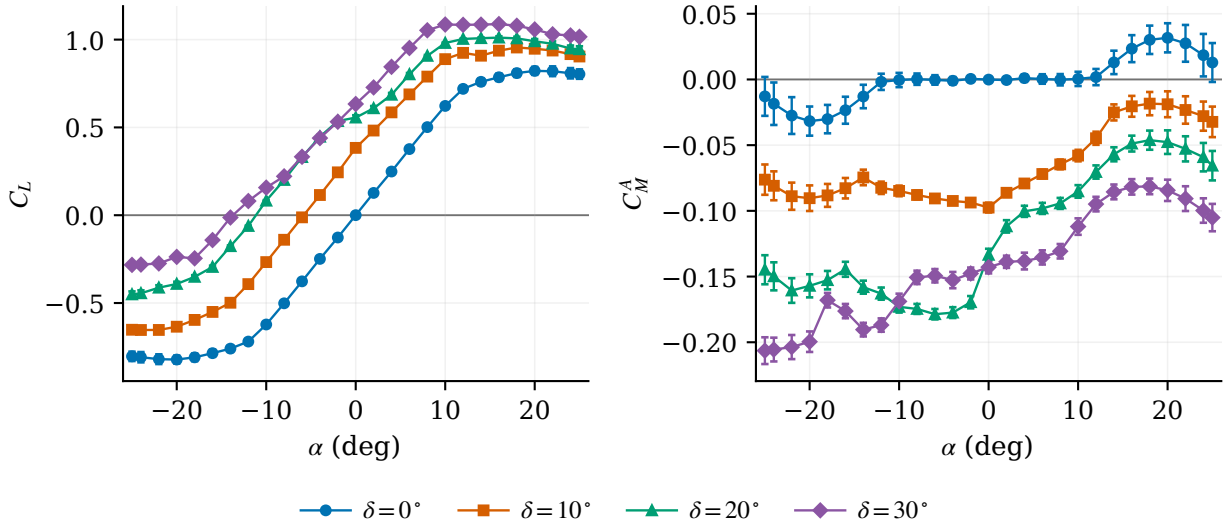



**Fig. 4 Experimental lift and pitching moment of the reference planform at zero dihedral for four control-surface deflections.**

For configuration I at $\delta = 20°$, the lift-coefficient increase over $\alpha = -2 \rightarrow 0°$ is approximately 0.020, compared with 0.089 over the preceding $\alpha = -4 \rightarrow -2°$ interval. This is a roughly 78% reduction in the interval's lift increase, while lift itself continues rising. Moment about the corresponding experimental aerodynamic center increases by +0.036, becoming less negative. Both departures in Table 3, calculated against $\alpha \in \{-6, -4, -2\}°$, exceed the supplied-uncertainty sensitivities. The pointwise lift and moment uncertainty magnitudes are approximately 0.013 and 0.0046–0.0047, respectively; the local incidence spacing is $\Delta\alpha = 2°$.

The two coefficients already respond differently at the lower deflection of $\delta = 10°$: the measured moment change is −0.0038 over $\alpha = -2 \rightarrow 0°$ and +0.011 over $\alpha = 0 \rightarrow 2°$, below the deflection giving the pronounced $\delta = 20°$ lift

**Table 3 Reference zero-dihedral measured lift and moment departures at $\delta = 20°$, with their uncertainty sensitivities.**

| Geom., $\Gamma$, $\delta$ | Coefficient | $\alpha$ interval (deg) | $D$ | $u_{\text{ind}}$ | $u_{\text{L1}}$ |
|---|---|---|---|---|---|
| I, 0°, 20° | $C_L$ | $-2 \to 0$ | -0.08274 | 0.02357 | 0.03631 |
| I, 0°, 20° | $C_M^A$ | $-2 \to 0$ | +0.03148 | 0.00862 | 0.01360 |

slowdown. At $\delta = 30°$, the early lift feature is broader. The lift-coefficient increases are 0.077, 0.064, and 0.111 over $\alpha = -12 \to -10°$, $\alpha = -10 \to -8°$, and $\alpha = -8 \to -6°$, while moment increases by about 0.018 on each of the first two intervals. Curvature of the preceding lift sequence makes the departure sensitive to the fitting range, so these actual interval changes describe the early feature more directly.

The numerical comparisons now assess separately the lift and moment evidence available for local interpretation. In Fig. 5, filled/solid and open/dashed curves denote WT and RANS, circles and squares distinguish $\delta = 20°$ and $\delta = 30°$ deflections, and open diamonds mark zero-incidence numerical values. Grey bands mark selected measured finite incidence intervals, plain for $\delta = 20°$ and hatched for $\delta = 30°$. They are neither uncertainty bands nor exact separation-onset locations. Each dataset's moment is about its corresponding aerodynamic center.

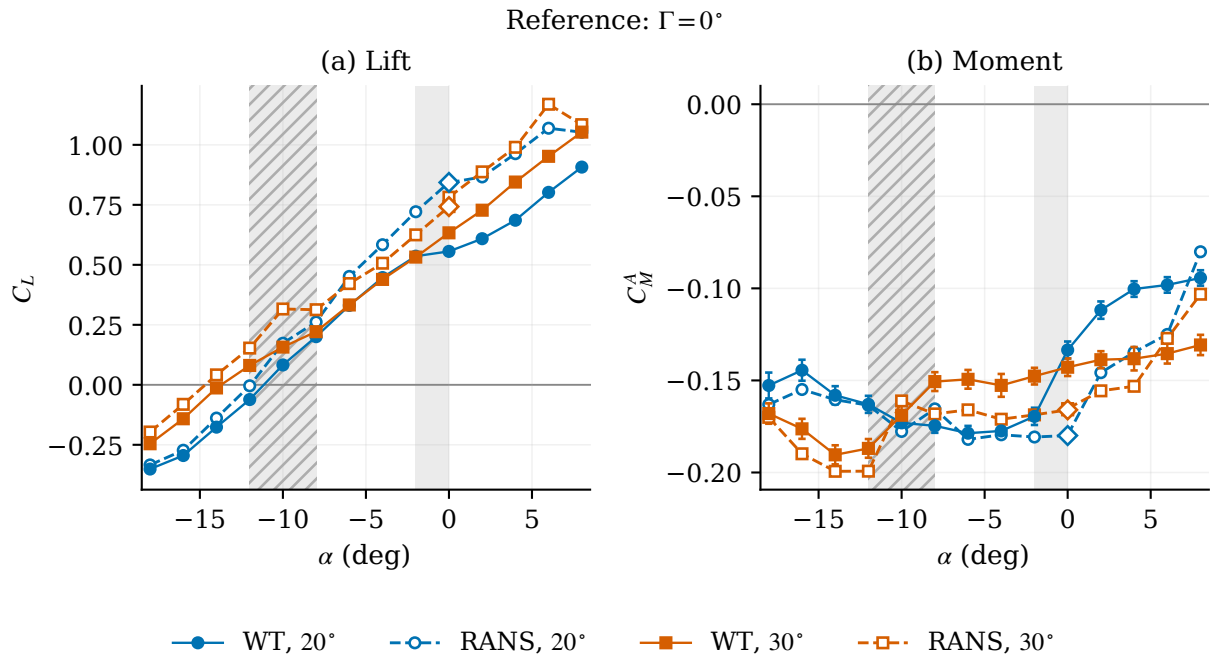



**Fig. 5 Reference at zero dihedral: WT and RANS lift and pitching moment. Grey bands: measured intervals, plain $\delta = 20°$, hatched $\delta = 30°$.**

For the reference model at $\delta = 20°$, the computed lift slowdown occurs over $\alpha = 0 \to 2°$, one sampled interval later than in WT. The numerical zero-incidence lift is approximately 0.842–0.843, compared with WT 0.556 and uncertainty magnitude 0.013. A qualitatively similar slowdown and less-negative moment thus coexist with a substantial lift-level

bias. At $\delta = 30°$, the early numerical lift change is sharper than the broad measured feature, and the moment departure depends more strongly on the preceding fitting range.

The reference computation captures the direction of the principal joint response. The comparisons with different geometries are presented with their loading analysis in Sec. V. The following reference analysis investigates the computed loading changes themselves, using the integrated comparison to delimit their experimental interpretation.

## IV. Joint Numerical Loading Response

### A. Fixed-surface and control-surface loads

The reference analysis concerns configuration I at zero dihedral and $\delta = 20°$. In Fig. 6, the upper panels show the fixed, control and total coefficients. Dashed segments continue each preceding fitted slope, anchored at its actual direct zero-incidence value, across the shaded $\alpha = 0 \rightarrow 2°$ interval. The endpoint gap is the departure; the lower signed markers distinguish the actual increment from that departure. The two displays answer different questions: whether a surface gains load, and how its growth differs from the preceding trend.

The fixed surface actually gains +0.0593 in lift coefficient, while the control loses −0.0351. Their combination leaves a positive total increase of +0.0242. Continued growth of total lift is therefore sustained by the fixed-surface gain exceeding the control loss. Nevertheless, the fixed-surface gain is smaller than expected from its preceding slope, and both surfaces contribute negative departures: −0.0692 fixed and −0.0414 control, with total −0.1106 (Table 4). The slowdown relative to the earlier growth is consequently broader than the actual control-surface lift loss.

The fixed-surface allocation of approximately 63% is the ratio $D_{L,\text{fixed}}/D_{L,\text{total}}$, using these two negative departures. It measures each surface's share of the computed lift-growth shortfall. Across the tested preceding-incidence ranges, this allocation remains 62.4–62.6%, while the total lift departure varies from approximately −0.105 to −0.137. The amplitude depends on the empirical preceding trend, but its division between surfaces is comparatively stable.

**Table 4 Reference zero-dihedral lift and aerodynamic-center moment departures from preceding incidence trends; total includes tip loads.**

| Geom., $\delta$, $\alpha$ interval | Departure | Fixed | Control | Total |
|---|---|---|---|---|
| I, 20°, $0 \rightarrow 2°$ | $D_L$ | -0.0692 | -0.0414 | -0.1106 |
| I, 20°, $0 \rightarrow 2°$ | $D_M^A$ | +0.0107 | +0.0226 | +0.0332 |
| I, 30°, $-10 \rightarrow -8°$ | $D_L$ | -0.0673 | -0.0529 | -0.1202 |
| I, 30°, $-10 \rightarrow -8°$ | $D_M^A$ | -0.0173 | +0.0150 | -0.0023 |

Pitching moment about the numerical aerodynamic center has a different component ranking. The departures reinforce: +0.0107 from the fixed surface and +0.0226 from the control. The fixed surface supplies the larger lift-growth deficit, but the control supplies the larger moment departure. The ranking depends on the axis: at the original numerical origin, fixed/control moment departures are +0.0529/+0.0474, so the fixed contribution is larger. Reinforcement persists

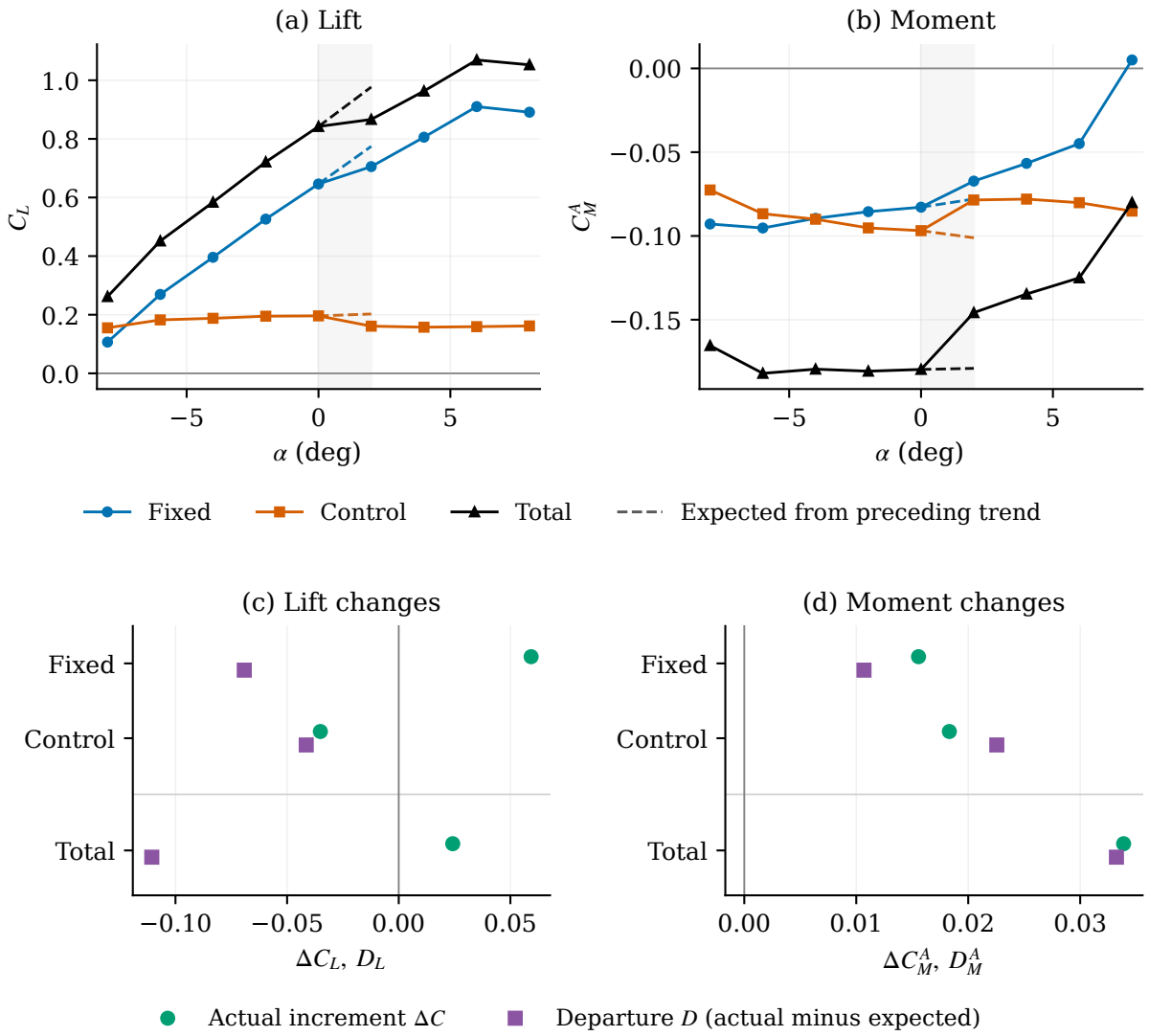



**Fig. 6 Fixed-surface, control-surface, and total numerical loading, with ordinary increments and departures from preceding incidence trends.**

at both axes. Locating these changes along the chord and span is therefore necessary to understand the combined force and moment response.

Tip loads contribute less than 0.09% of the changes listed in Table 4 and are included in the totals. The dihedral budget retains their appreciable contribution explicitly.

### B. Pressure distribution and pitching moment

Figure 7 first resolves the local pressure change between the actual $\alpha = 0°$ and $\alpha = 2°$ states of the reference budget. Upper and lower sides are overlaid: solid curves with filled markers denote the upper side, and dotted curves with open markers denote the lower side. Circles and squares distinguish incidence. Upper leading-edge suction strengthens,

while the suction peak near the hinge weakens. Here, pressure recovery denotes the chordwise rise downstream of the hinge suction minimum. The change in pressure recovery, together with the stronger leading-edge suction, redistributes loading along the chord.

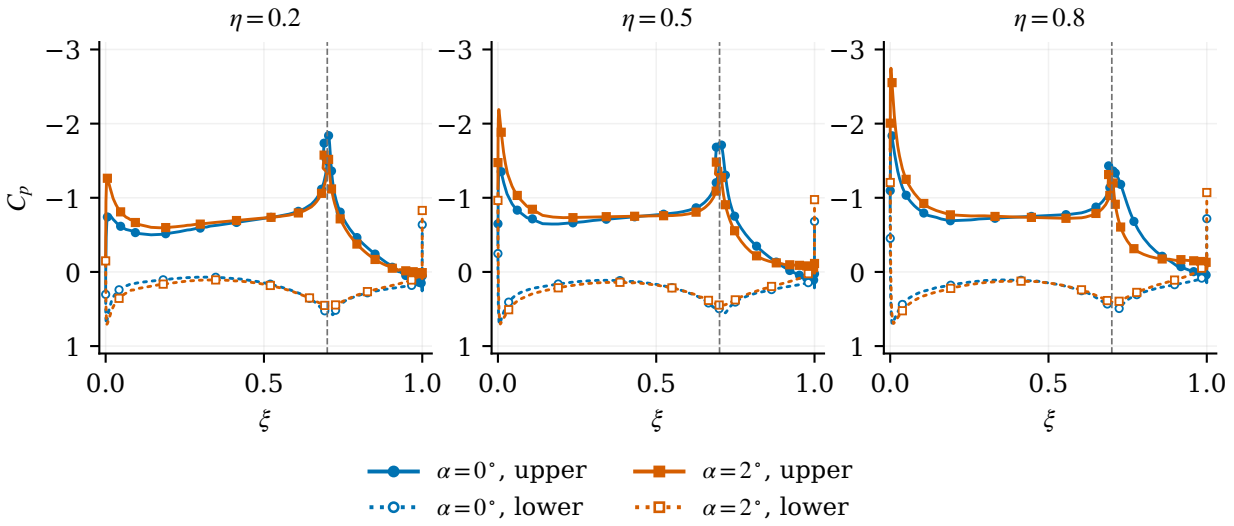



**Fig. 7 Chordwise pressure distributions at three span locations across the reference pre-stall loading change.**

Figure 8(a) makes the connection to growth by integrating each chord region over the semispan and comparing actual with expected lift increments. The forward region gains +0.0671, consistent with stronger leading-edge suction, but its preceding slope predicts +0.1027; its departure is −0.0357. The intermediate fixed-chord region instead loses −0.0097 rather than gaining the expected +0.0257, giving a departure of −0.0354. Stronger suction near the leading edge therefore coexists with both reduced forward-region growth and an actual intermediate-region loss. Their combination explains why the complete fixed surface still gains lift while contributing strongly to the slowdown in Fig. 6.

The aft region loses −0.0332, compared with an expected +0.0063, giving a departure of −0.0396. Pressure supplies nearly all the principal lift departure, while viscous forces remain included in the integration.

Panels (b,c) answer a different question from panel (a): they show actual aft-region ($\xi \geq 0.7$) lift and moment increments in three integrated spanwise bins, with no preceding-trend subtraction. The aft lift changes are similar, −0.0103, −0.0118, and −0.0111 from inboard to outboard, yet moment changes increase from +0.00252 through +0.00612 to +0.00910. The aft-swept geometry in Fig. 3 places outboard aft loading farther downstream of the numerical aerodynamic center. For a local decrease in upward body-normal force, the factor $-(x - x_A)\, dF_z$ gives a positive moment change, larger for a longer downstream arm. The integrated body-normal force decreases in all three aft bins, supporting this interpretation; their detailed chordwise redistribution also enters the integral. The $F_x$ moment term $z\, dF_x$ remains included and is small here. The longer outboard arms therefore turn comparable aft lift losses into progressively larger positive moment increments.

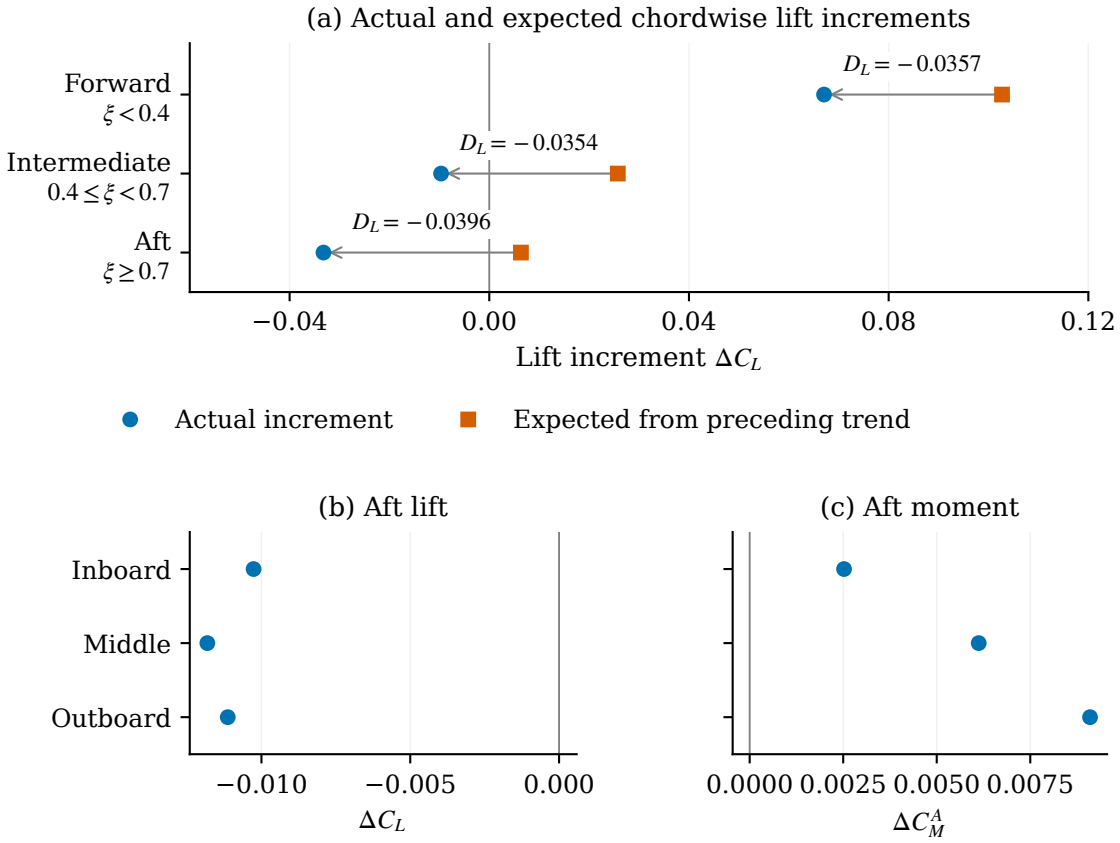



**Fig. 8 Chordwise and spanwise contributions to reference lift and pitching-moment changes over the selected pre-stall interval.**

### C. Surface-flow evolution

The pressure and load changes can now be compared with the evolution of near-wall flow on the deflected control. Figure 9 reports the proportion of its upper surface on which chordwise wall shear points upstream, using the definition in Sec. II.D. Only the primary $C_{f,c} < 0$ measure is plotted. The open diamond marks the $\alpha = 0°$ solution used in the surface maps and load analysis.

For the reference $\delta = 20°$ calculation, that area fraction grows from about 15% to 49% (0.146 to 0.485) across $\alpha = 0 \rightarrow 2°$, the same interval as the reduced fixed-surface growth, control lift loss and positive moment change. The maps in Fig. 10 distinguish this increase in area from a change in location.

Figure 10 shows the signed coefficient $C_{f,c}$ on the upper surface for both reference deflections, using identical projections and color scales. At $\delta = 20°$, the $\alpha = 0°$ state has a narrow reversed-shear band adjacent to the hinge, followed by predominantly downstream shear. At $\alpha = 2°$, the dominant negative-shear region instead lies near the trailing edge. At $\alpha = 4°$, this trailing-edge-region reversal extends farther upstream. The change across $\alpha = 0 \rightarrow 2°$ therefore includes a reorganization of the reversal pattern, alongside the weakened hinge suction and aft load loss in

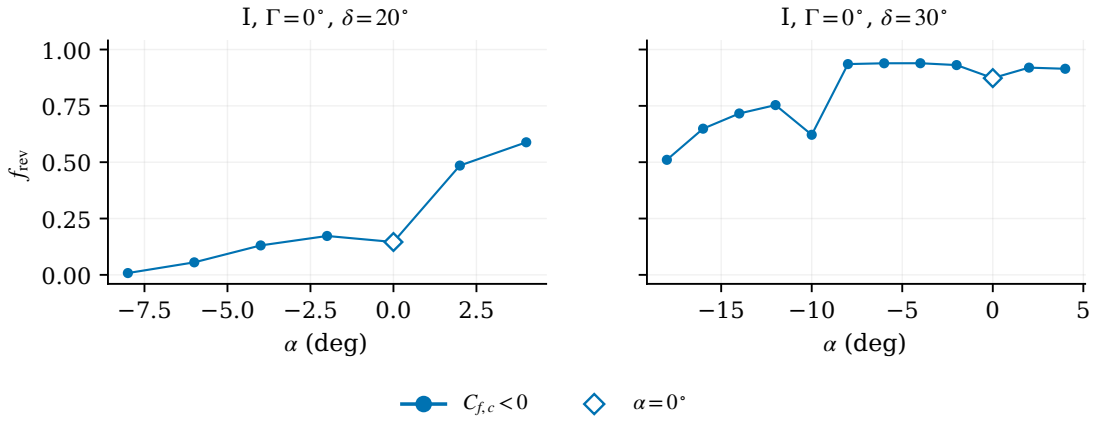



**Fig. 9 Reference zero-dihedral upper-control area fraction with reversed chordwise wall shear at $\delta = 20°$ and $\delta = 30°$.**

Figs. 7 and 8; it is not simply uniform growth of the initial hinge-adjacent band.

By $\alpha = 4°$, about 59% of the upper control has reversed chordwise shear, while the total lift increase over $\alpha = 2 \rightarrow 4°$ is 0.0968, larger than the preceding 0.0242. Stronger total lift growth can thus coexist with a larger reversed area because the integrated response combines the evolving loading on both surfaces. Reversal extent characterizes near-wall flow, but does not by itself determine the pressure force or pitching moment.

At $\delta = 30°$, the reference lift slowdown occurs much earlier, over $\alpha = -10 \rightarrow -8°$. Fixed lift still increases by +0.0406, while the control loses −0.0436, leaving a small total decrease of −0.0030. Both lift departures remain negative. The lower row of Fig. 10 follows this earlier interval. Over it, the upper-control reversed-shear fraction rises from approximately 62% to 94% (0.622 to 0.935). The recurring association is thus reduced aft loading with expansion of pre-existing reversal, although the total lift increment changes sign because the fixed gain no longer outweighs the control loss.

The moment result at $\delta = 30°$ is different: the fixed-surface departure is negative and the control departure positive, so they partly cancel (Table 4). Across the tested preceding fitting ranges, total moment departure varies from approximately −0.0262 to +0.0076 as the opposing contributions change their balance. Its total sign is therefore not robust to this empirical reference. This contrasts with the reinforcing moment departures at $\delta = 20°$ and shows why reversed-flow area alone is insufficient to infer the sign of the integrated pitching response.

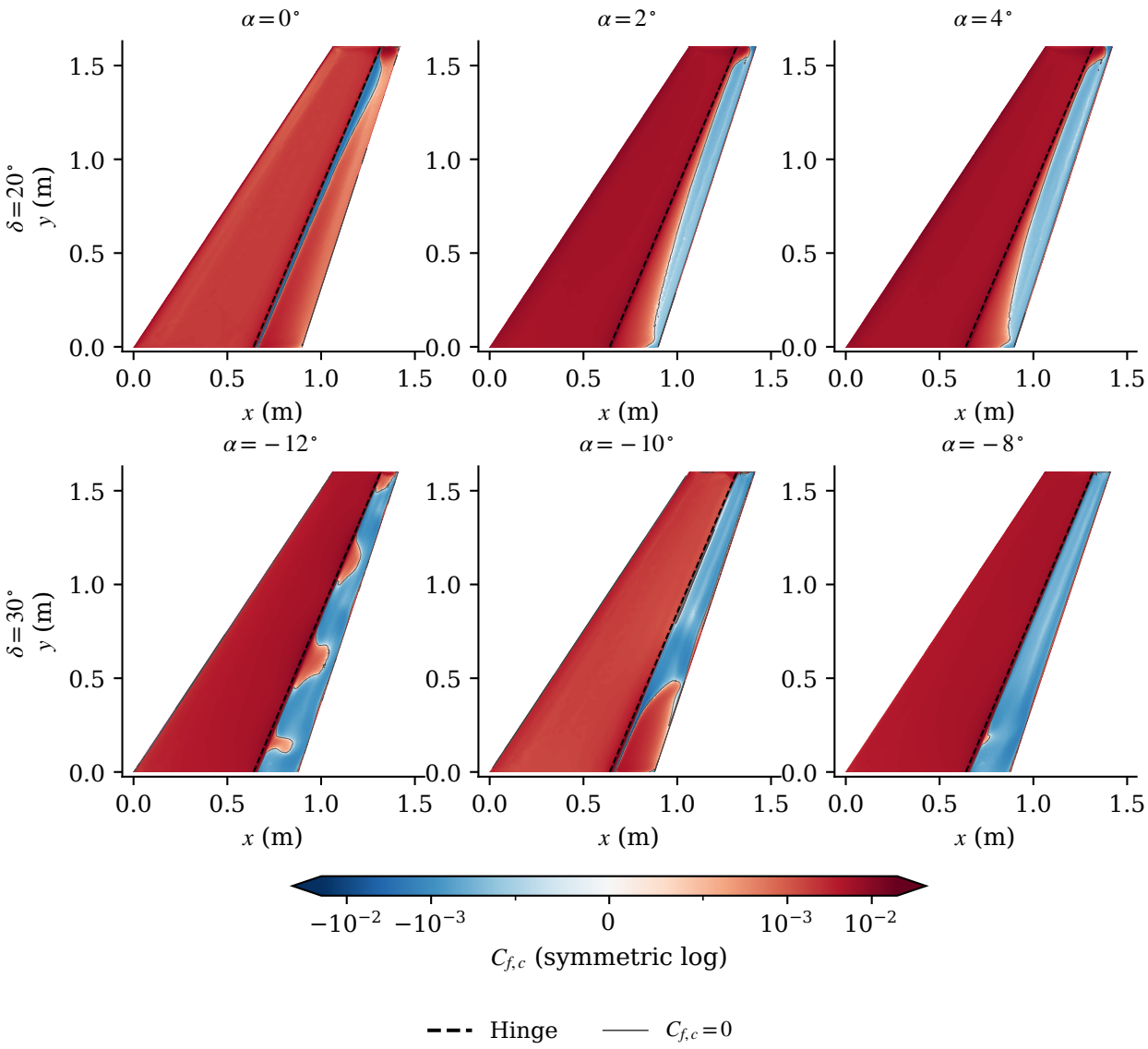



**Fig. 10 Reference upper-surface chordwise skin friction across the loading changes at $\delta = 20°$ (top) and $\delta = 30°$ (bottom) deflection.**

## V. Dihedral and Planform Dependence

The same component and spatial accounting is now applied to the dihedral and planform comparisons. Their measured responses and WT–RANS correspondence are presented beside the numerical loading analysis.

Table 5 reports the measured departures and uncertainty sensitivities for these geometry comparisons, using the definitions in Sec. II.D.

**Table 5 Measured lift and moment departures for the dihedral and forward-swept cases, with their uncertainty sensitivities.**

| Geom., $\Gamma$, $\delta$ | Coefficient | $\alpha$ interval (deg) | $D$ | $u_{\text{ind}}$ | $u_{\text{L1}}$ |
|---|---|---|---|---|---|
| I, 45°, 20° | $C_L$ | $-2 \rightarrow 0$ | -0.04785 | 0.01694 | 0.02634 |
| I, 45°, 20° | $C_M^A$ | $-2 \rightarrow 0$ | +0.00847 | 0.02630 | 0.04082 |
| I, 45°, 30° | $C_L$ | $-2 \rightarrow 0$ | -0.04038 | 0.01646 | 0.02996 |
| I, 45°, 30° | $C_M^A$ | $-4 \rightarrow -2$ | -0.02860 | 0.02478 | 0.03822 |
| II, 0°, 20° | $C_L$ | $0 \rightarrow 2$ | -0.06902 | 0.02090 | 0.03739 |
| II, 0°, 20° | $C_M^A$ | $0 \rightarrow 2$ | +0.01668 | 0.00793 | 0.01424 |

### A. Dihedral

Figure 11 brings together the measured dihedral contrast (upper panels) and the $\Gamma = 45°$ WT–RANS comparison (lower panels). At $\delta = 20°$, measured lift growth remains suppressed across $\alpha = -2 \rightarrow 2°$, while the moment departure over $\alpha = -2 \rightarrow 0°$ is smaller than both uncertainty sensitivities in Table 5. At $\delta = 30°$, the selected lift slowdown occurs over $\alpha = -2 \rightarrow 0°$, but moment decreases earlier, by 0.027 over $\alpha = -4 \rightarrow -2°$. That moment departure is comparable with $u_{\text{ind}}$ and smaller than $u_{\text{L1}}$. The joint response differs from the clearly resolved reference $\delta = 20°$ lift/moment change.

The computed sequence has weaker correspondence with these measurements. At $\delta = 20°$, numerical lift decreases over $\alpha = -6 \rightarrow -4°$ while measured lift increases; RANS does not convincingly reproduce the sustained measured slowdown or its weak moment feature. At $\delta = 30°$, the negative numerical moment change occurs over $\alpha = -2 \rightarrow 0°$, one sampled interval later than the measured change. The following decomposition consequently concerns the computed dihedral response.

Lift is defined in the same global frame for both geometries. At $\alpha = 0°$, it is the vertical force $F_z$. The projection in Fig. 12(a) concerns the force normal to the panel, denoted $F_n$, whose direction tilts with dihedral. A force of unchanged magnitude would then provide only $F_n \cos\Gamma$ vertically. A force $F_s$ along the tilted span also contributes vertically, giving $F_z = F_n \cos\Gamma + F_s \sin\Gamma$.

Figure 12(b) illustrates this distinction at $\alpha = 0°$, $\delta = 20°$. Tilting the reference panel-normal force while holding its magnitude fixed gives $0.8422 \cos 45° = 0.5955$ in lift coefficient. This accounts for 0.2467 of the reduction from the reference value. The actual $\Gamma = 45°$ result is 0.5407; the remaining reduction of 0.0548 comes from the changed panel-normal load and the vertical contribution of the local spanwise force. The lift direction remains fixed throughout this comparison.

A constant scaling of lift cannot describe the computed incidence response. At the same $\delta = 20°$ deflection, the actual lift increase over $\alpha = -2 \rightarrow 0°$ is 0.1210 at zero dihedral and 0.0763 at $\Gamma = 45°$. Across the next interval, $\alpha = 0 \rightarrow 2°$, the ordering reverses: the increases are 0.0242 and 0.0444. Although the dihedral model has lower absolute lift, it gains more lift over that latter interval. The change in ordering establishes an incidence-dependent

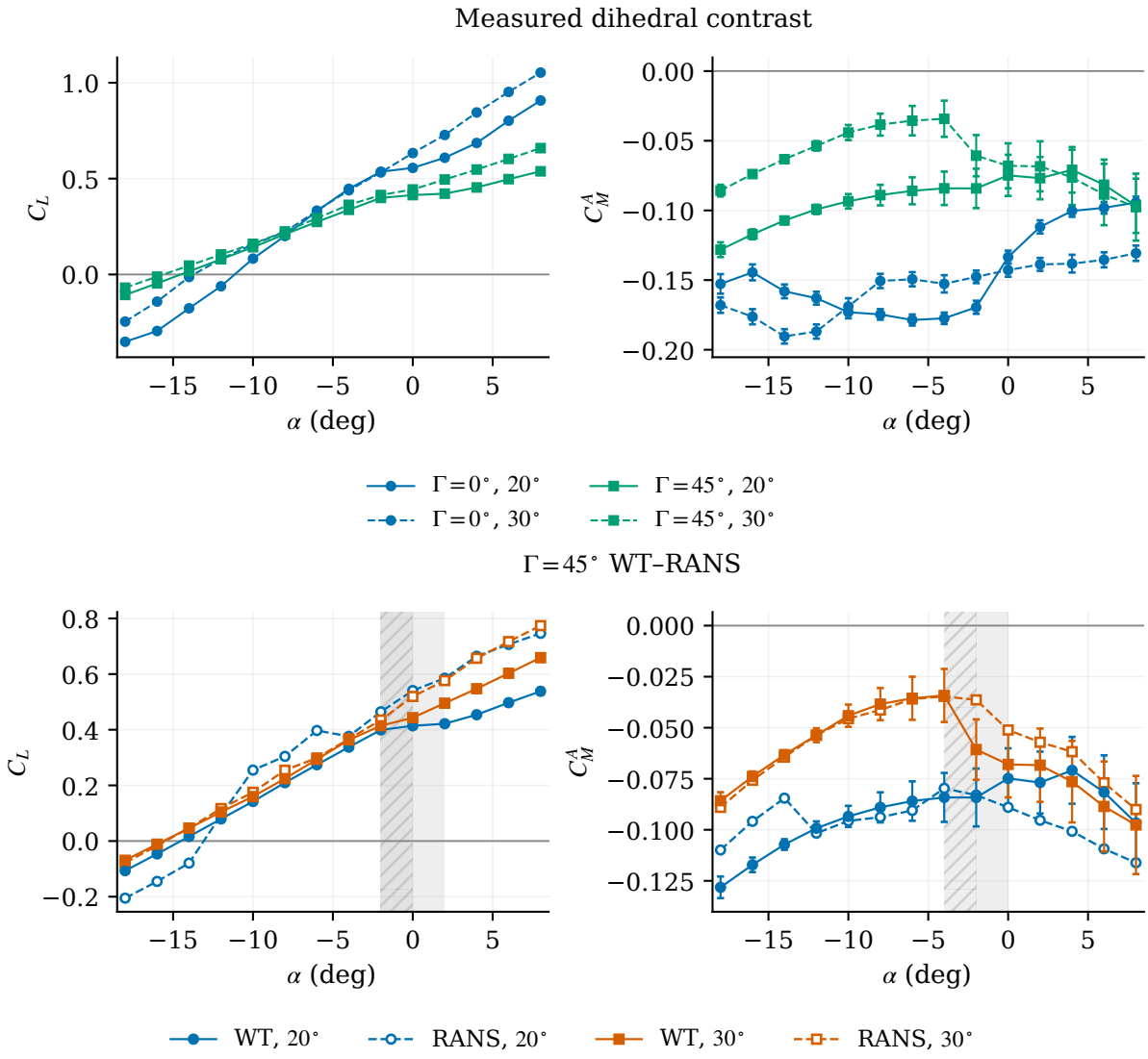



**Fig. 11 Measured dihedral contrast (top) and positive-dihedral WT–RANS comparison (bottom). Grey bands mark measured intervals: plain $\delta = 20°$, hatched $\delta = 30°$.**

difference in computed loading beyond a fixed projection factor. Table 6 shows that the $\Gamma = 45°$ fixed surface still gains lift over $\alpha = 0 \rightarrow 2°$, while the control loses lift. Its moment about the numerical aerodynamic center decreases through contributions from the fixed surface, control and tip. The table retains the tip explicitly because its moment contribution is appreciable in these dihedral cases.

Pitching moment depends on where $F_x$ and $F_z$ act. Dihedral places the loading above the reference axis, so $F_x$ contributes through the vertical arm $z$. The $F_z$ contribution is weighted by the downstream arm $x - x_A$ (Eq. 5). Both

**Table 6 Actual coefficient increments for the dihedral and forward-swept cases; moments are about their corresponding numerical aerodynamic centers.**

| Geom., $\Gamma$, $\delta$, $\alpha$ interval | Increment | Fixed | Control | Tip | Total |
|---|---|---|---|---|---|
| I, 45°, 20°, $0 \rightarrow 2°$ | $\Delta C_L$ | +0.0532 | -0.0106 | +0.0018 | +0.0444 |
| I, 45°, 20°, $0 \rightarrow 2°$ | $\Delta C_M^A$ | -0.0044 | -0.0003 | -0.0017 | -0.0063 |
| I, 45°, 30°, $-2 \rightarrow 0°$ | $\Delta C_L$ | +0.0864 | -0.0028 | +0.0013 | +0.0848 |
| I, 45°, 30°, $-2 \rightarrow 0°$ | $\Delta C_M^A$ | -0.0112 | -0.0025 | -0.0011 | -0.0148 |
| II, 0°, 20°, $0 \rightarrow 2°$ | $\Delta C_L$ | +0.1014 | -0.0079 | 0.0000 | +0.0935 |
| II, 0°, 20°, $0 \rightarrow 2°$ | $\Delta C_M^A$ | +0.0112 | +0.0057 | 0.0000 | +0.0168 |

numerical solutions are referred to the same physical axis $A_I$. Figure 12(c,d) compares their moment increments through

$$\mathcal{I}_M = [C_{M,45}^{A_I}(\alpha_b) - C_{M,45}^{A_I}(\alpha_a)] - [C_{M,0}^{A_I}(\alpha_b) - C_{M,0}^{A_I}(\alpha_a)] \tag{6}$$

A negative $\mathcal{I}_M$ means that the moment increase is smaller, or the decrease larger, with dihedral over the same incidence interval. These are differences of actual moment increments, not departures from preceding fitted trends.

At $\delta = 20°$, the $F_x$ moment term gives a negative contrast in every plotted interval. Over $\alpha = -4 \rightarrow -2°$, it contributes −0.00362, partly offset by a +0.00131 contribution from $F_z$, leaving $\mathcal{I}_M = -0.00231$. Across $\alpha = 0 \rightarrow 2°$, the terms instead reinforce: −0.01283 from $F_x$ and −0.02750 from $F_z$ give −0.04033. The $F_z$ moment term supplies the larger difference during this reference loading change. Over $\alpha = 2 \rightarrow 4°$, its contribution falls to −0.00260, while the contribution from $F_x$ remains −0.01412 and dominates the −0.01672 total. A single relative weighting of the two moment terms therefore cannot describe the sequence.

At $\delta = 30°$, the terms reinforce over $\alpha = -2 \rightarrow 0°$: −0.00808 from $F_x$ and −0.01027 from $F_z$ give a total −0.01836. Over $\alpha = 0 \rightarrow 2°$, both remain negative, with the contribution from $F_x$ larger. Over $\alpha = 2 \rightarrow 4°$, a positive contrast from $F_z$ of +0.00556 offsets part of the −0.01286 contrast from $F_x$, leaving −0.00730. This change from reinforcement to compensation concerns the moment-term differences between geometries; it is distinct from fixed/control moment-departure reinforcement in Sec. IV.A.

The upper control already has widespread reversed flow throughout the interval in Fig. 13: the reversed-area fraction is 0.889 at $\alpha = -2°$, 0.911 at $\alpha = 0°$ and 0.912 at $\alpha = 2°$. The moment decrease therefore occurs with little change in the extent of control-surface reversal.

Figure 14 relates this extensive control reversal to the changing flow over the fixed surface. Between $\alpha = -2°$ and $\alpha = 0°$, a band of increased chordwise shear develops from approximately midspan toward the tip. The accompanying rise in the algebraic transition-model factor $\gamma_{\mathrm{BC}}$ identifies activation of modeled turbulence production in this region. At $\alpha = 2°$, a narrow leading-edge reversed-shear patch appears over roughly the outer third of the span. At the $\eta = 0.8$ cut, negative shear with small $\gamma_{\mathrm{BC}}$ is followed downstream by positive shear and increasing $\gamma_{\mathrm{BC}}$. This sequence is consistent

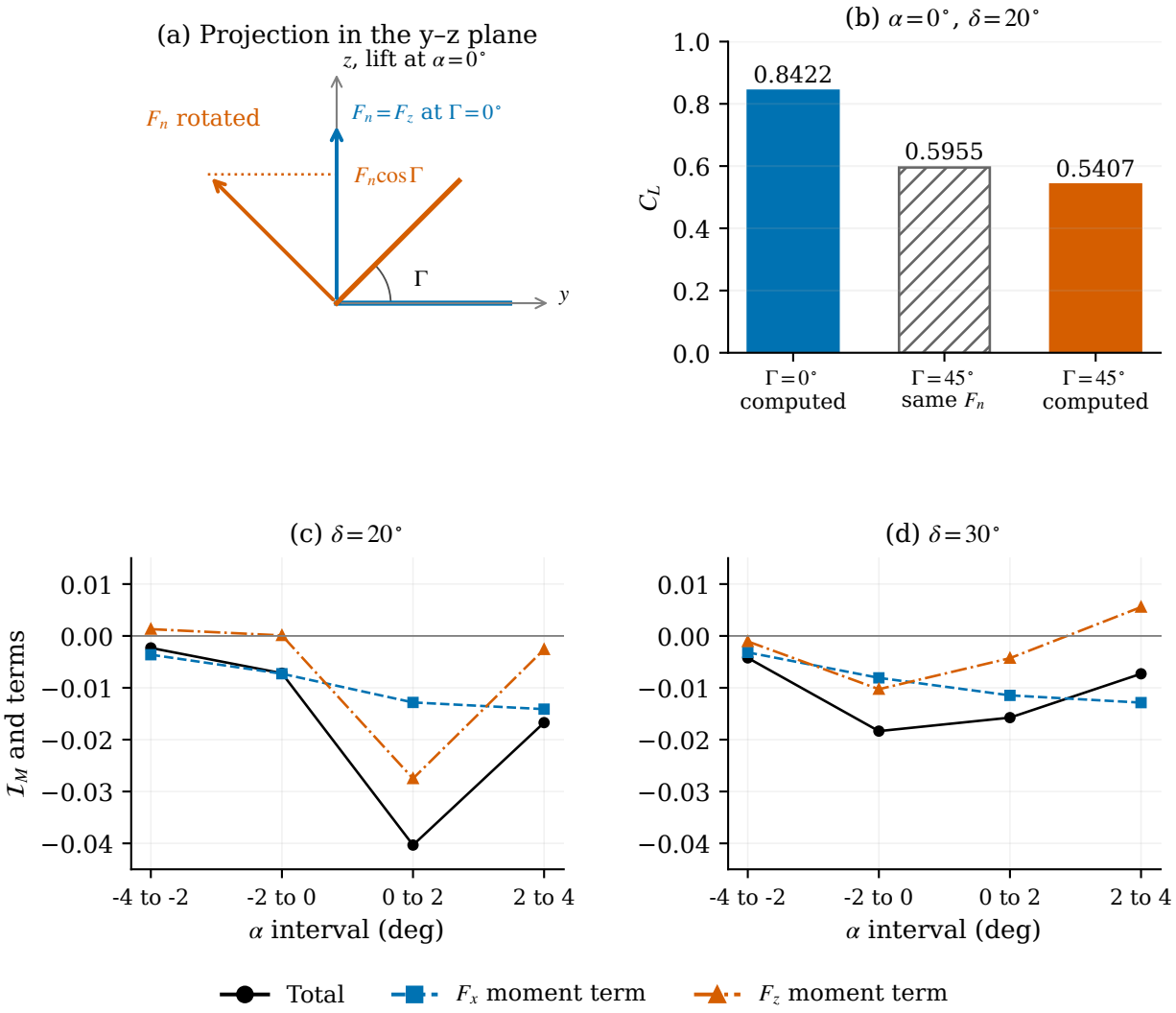



**Fig. 12 Panel-normal force projection and computed lift (top); dihedral differences in moment increments from $F_x$ and $F_z$ (bottom).**

with a model-predicted laminar separation bubble followed by transition and recovery of downstream wall shear.

The pressure sections show the accompanying spanwise variation of loading. At $\alpha = 0°$, the leading-edge suction is weak at $\eta = 0.2$, whereas distinct peaks appear at $\eta = 0.5$ and 0.8; their approximate minima are $C_p = -0.52, -1.21, -1.63$. The hinge-region suction also changes with span, with stronger peaks at the middle and outboard cuts. Stronger suction followed by pressure recovery gives the outboard boundary layer a different pressure-gradient history, consistent with the transition and leading-edge reversal developing there first. Spanwise variation of local inflow provides a possible contribution to the suction difference. Taper reduces local chord and Reynolds number, and the constant section shape gives a smaller dimensional leading-edge radius outboard.

Over most of the control chord, the pressure curves change little between $\alpha = -2°$ and $\alpha = 2°$, consistent with the

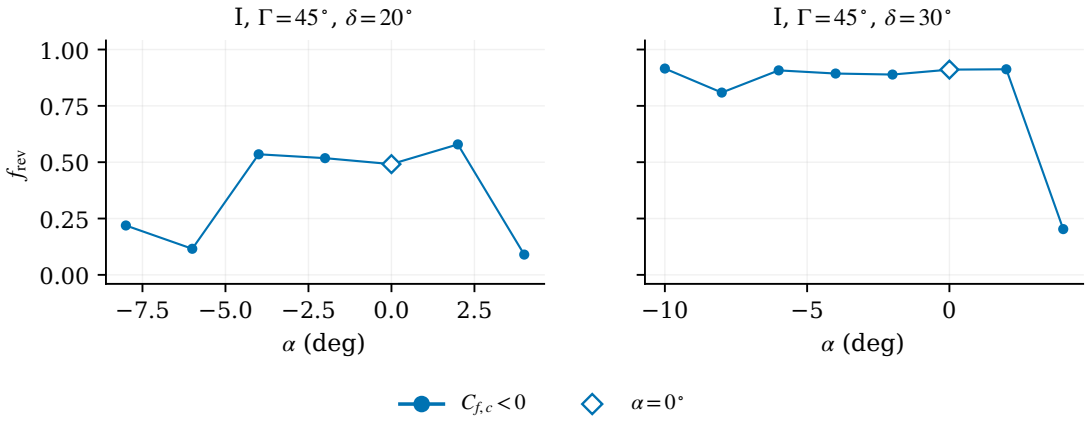



**Fig. 13 Upper-control reversed-shear fractions for $\Gamma = 45°$ at $\delta = 20°$ and $\delta = 30°$.**

nearly unchanged reversed area. The fixed-surface shear and pressure changes therefore occur while the control remains extensively separated. The component increments in Table 6 and the $F_x/F_z$ terms in Fig. 12(c,d) show how this loading produces the computed moment decrease.

Dihedral changes the vertical component of the panel loading and gives $F_x$ a vertical moment arm. As incidence changes, the contributions from $F_x$ and $F_z$ reinforce or partly offset one another, producing the changing moment response in Fig. 12(c,d).

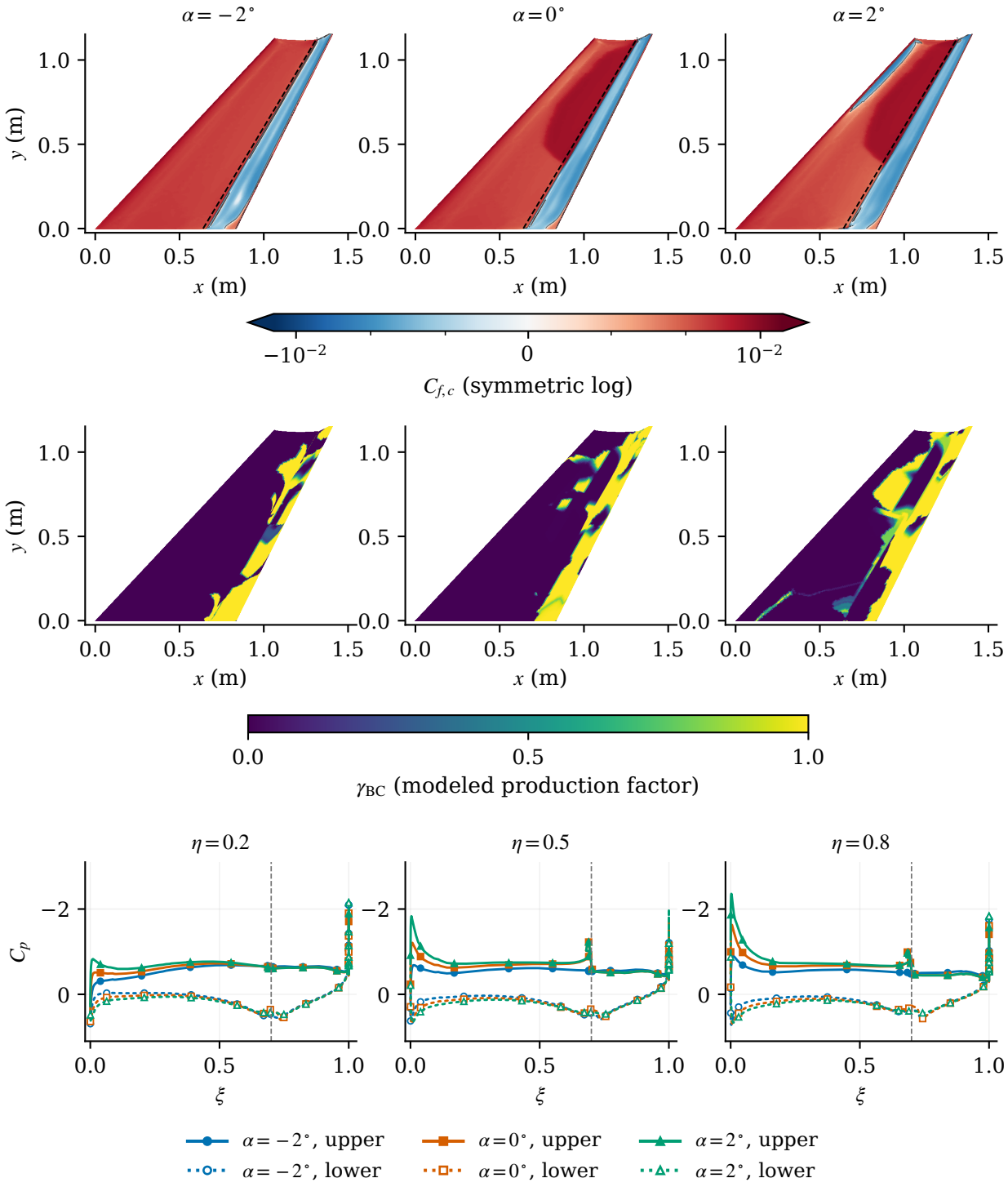



**Fig. 14** $\Gamma = 45°$, $\delta = 30°$**: upper-surface chordwise shear (top), transition-model factor (middle), and pressure sections (bottom) at** $\alpha = -2, 0, 2°$**.**

**B. Aft- and forward-swept planforms**

The second comparison reverses quarter-chord sweep from $\lambda = +30^\circ$ to $\lambda = -15^\circ$ while retaining the 1.6 m semispan and full-span 30%-chord control. Sweep is the principal geometric contrast. Area and MAC differ by approximately 4% and 8%, respectively, and taper changes from 0.4 to 0.6 (Table 1); these accompanying differences prevent attributing the response to sweep alone.

The wind-tunnel results in Fig. 15 show configuration II's local lift slowdown at $\delta = 20^\circ$ over $\alpha = 0 \rightarrow 2^\circ$: lift increases by 0.044, compared with approximately 0.092 over both $\alpha = -2 \rightarrow 0^\circ$ and $\alpha = 2 \rightarrow 4^\circ$. Moment increases by 0.019 over $\alpha = 0 \rightarrow 2^\circ$. This feature is later than on configuration I, with a smaller moment rise across its own interval. The primary moment departure in Table 5 exceeds both uncertainty sensitivities, although neighboring preceding fitting ranges do not uniformly exceed the primary L1 sensitivity. At $\delta = 30^\circ$, the reference model's broad early feature is not repeated at the same negative incidences by II.

RANS gives a direct $\alpha = 0 \rightarrow 2^\circ$ lift increase of 0.0935 at $\delta = 20^\circ$, more than twice the measured value. Its moment increase, 0.0168, is locally close to the measured 0.019, with the different aerodynamic-center positions retained. This weaker lift correspondence limits the use of its local solution to explain the measured feature.

Figure 16 compares direct $\alpha = 0 \rightarrow 2^\circ$ increments at $\Gamma = 0^\circ$, $\delta = 20^\circ$. Panel (a) shows that the forward-swept fixed surface gains +0.1014, compared with +0.0593 on configuration I, while its control loses only −0.0079 rather than −0.0351. The total increase is consequently +0.0935 instead of +0.0242. The same sign combination of fixed gain and control loss persists, but its balance is substantially different. Each planform uses its own reference area from Table 1.

Panel (b) locates the lift difference along the chord. The forward-region increase is +0.0938 for II versus +0.0671 for I. More significantly, the intermediate region gains +0.0063 on II while losing −0.0097 on I; the aft loss is −0.0065 versus −0.0332. The forward-swept model thus combines stronger forward loading with continued intermediate-chord growth and much weaker aft loss. Its fixed-surface actual gain is not simply a scaled version of the reference pressure redistribution.

Both models nevertheless fall below their own preceding trends. The forward/intermediate/aft lift departures are −0.0126/−0.0152/−0.0090 for II, compared with −0.0357/−0.0354/−0.0396 for I. Thus II's positive intermediate increment still represents reduced growth, whereas the lift contribution of I's intermediate region decreases. The fixed/control departures are −0.0263/−0.0104 for II versus −0.0692/−0.0414 for I. The weaker computed total deficit on II is therefore distributed across the chord, with the largest between-planform change in the aft region.

The paired rows separate lift from moment in every decomposition. Panels (a,d) show that I's fixed/control lift changes of +0.0593/−0.0351 accompany positive moment increments of +0.0156/+0.0183. On II, the fixed/control moment increments are +0.0112/+0.0057. Panels (b,e) locate the smaller total moment rise on II mainly in the intermediate and aft regions: their moment increments are +0.0018/+0.0048, compared with +0.0057/+0.0177 on I, while the forward-region increments are similar.

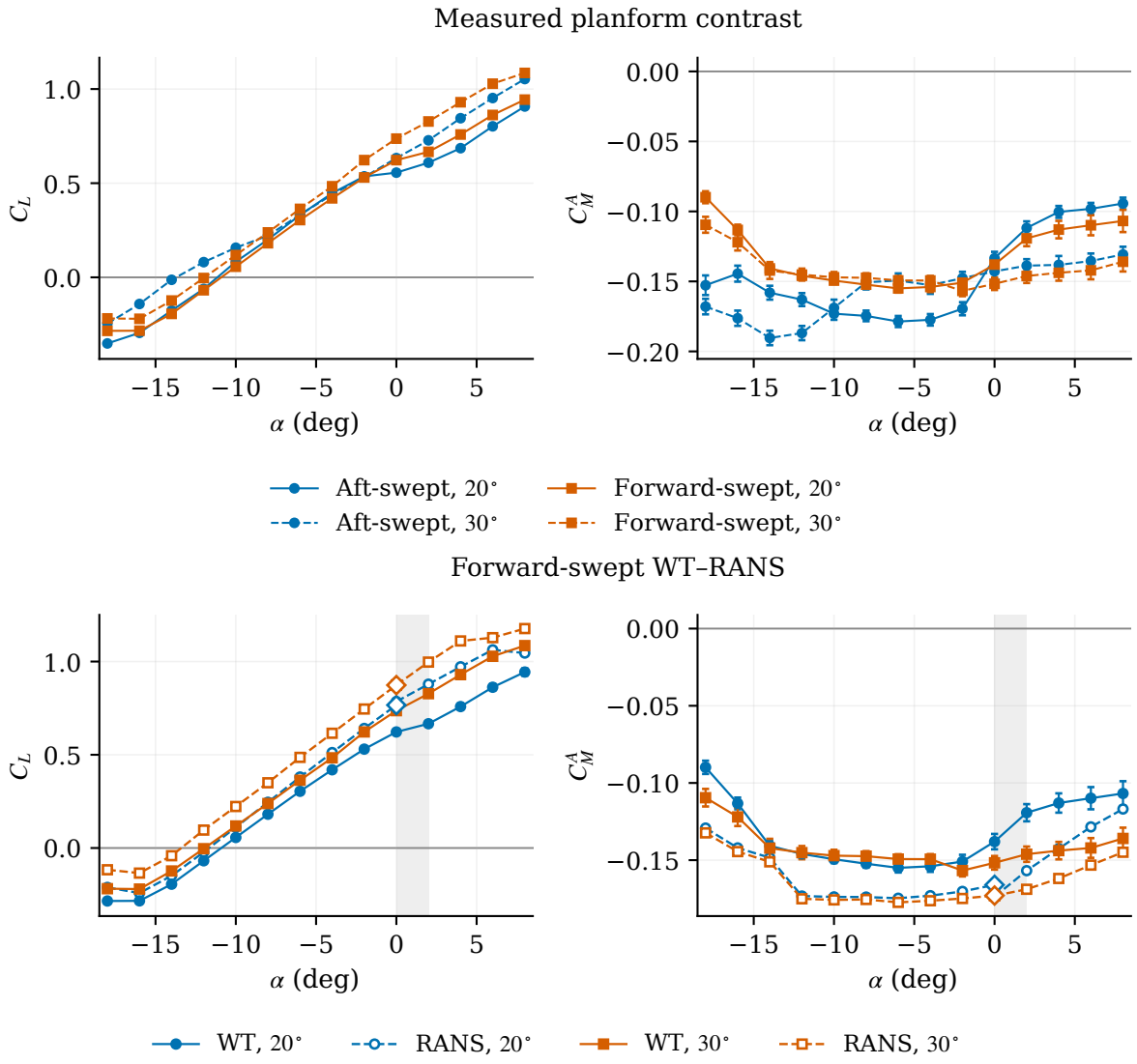



**Fig. 15 Measured planform contrast (top) and forward-swept WT–RANS comparison (bottom). Grey band marks the measured $\delta = 20°$ interval.**

Panels (c,f) then show how aft lift and moment changes vary along the span. For I, similar aft lift losses of $-0.0103, -0.0118, -0.0111$ give positive moment increments of $+0.0025, +0.0061, +0.0091$ from inboard to outboard. For II, the aft lift changes are $-0.0077, -0.0007, +0.0018$: the loss is concentrated inboard and the outboard region gains lift. Its corresponding moment increments are $+0.0054, -0.0002, -0.0004$.

Sweep places these aft loads differently relative to the moment axis. On I, the aft region lies progressively farther downstream toward the tip; on II, it lies progressively closer to the axis outboard. Both aft regions remain downstream of their moment axes, with the arm $x - x_A$ growing outboard on I and decreasing on II. A loss of upward force downstream of the axis gives a positive moment change through $-(x - x_A)dF_z$, with greater weight at a longer arm. This explains

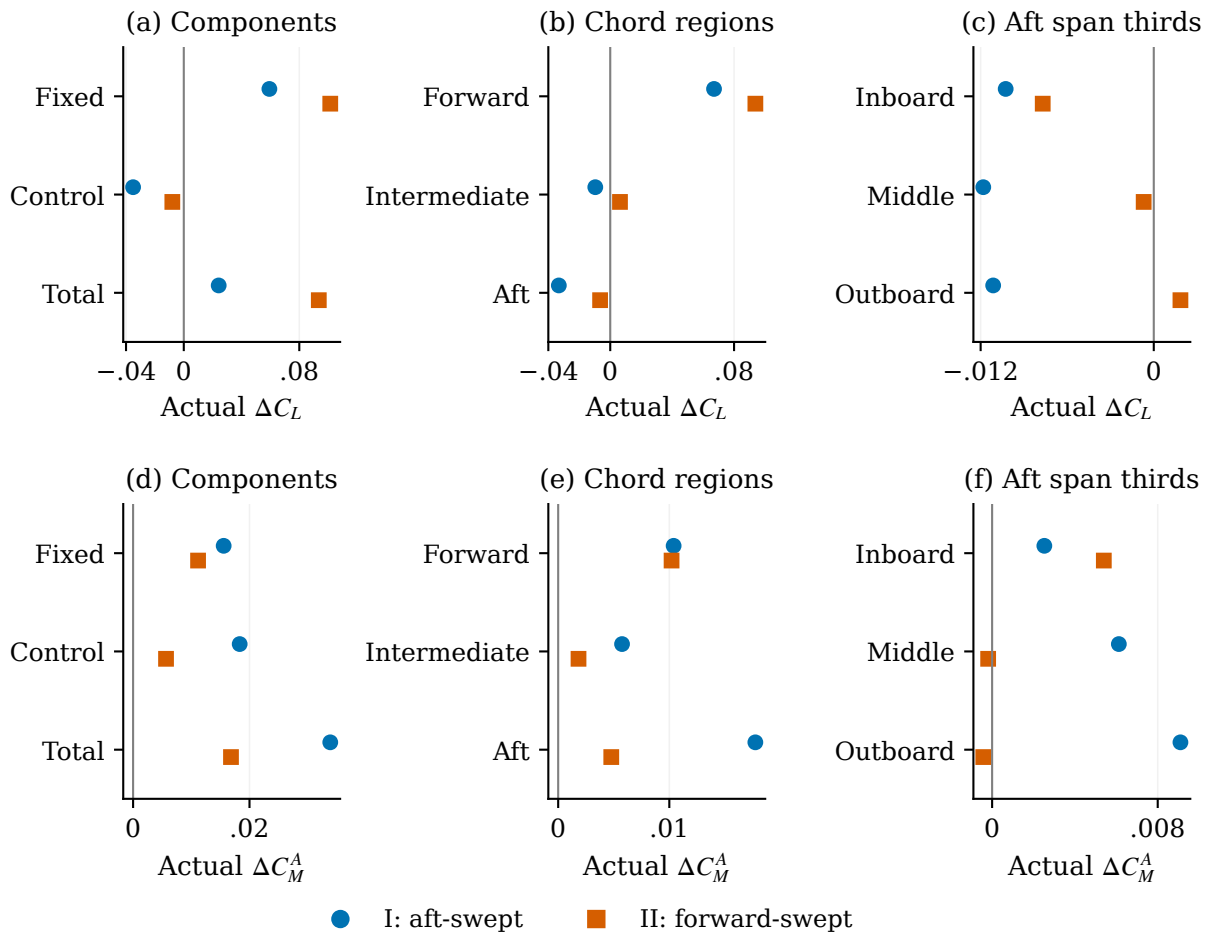



**Fig. 16 Planform comparison over $\alpha = 0 \rightarrow 2°$, $\delta = 20°$: lift increments (top) and aerodynamic-center moment increments (bottom), with matching decompositions.**

why I's comparable aft lift losses have increasing outboard moment contributions, whereas II's inboard-concentrated loss has its largest positive moment contribution inboard. The moment row includes the $F_x$ contribution as well as the $F_z$ contribution.

Moment departures reinforce in both planforms, although their component ranking changes. Configuration I gives +0.0107 fixed and +0.0226 control, whereas II gives approximately +0.0075 and +0.0071. The control dominates the reference moment departure; the two contributions are similar on II. This reinforcement is reference-dependent: at II's original numerical origin, the fixed/control departures are −0.0023/+0.0035 and partly cancel. Together with the chordwise and spanwise results, the aerodynamic-center comparison establishes a different computed organization of force growth and pitching response on the forward-swept model.

Figure 17 places II's reversed-area progression beside its loading analysis. The upper-control fraction grows from about 19% to 47% and 59% at $\alpha = 0, 2, 4°$. The surface maps in Fig. 18 locate this change: at $\alpha = 0°$, trailing-edge-region

reversal is concentrated around midspan; with increasing incidence it extends upstream along the control chord and toward the root. The pressure sections across $\alpha = 0 \rightarrow 2^\circ$ show a weakened upper-side hinge-suction peak inboard and comparatively little change at the outboard cut, consistent with the concentration of aft lift loss inboard.

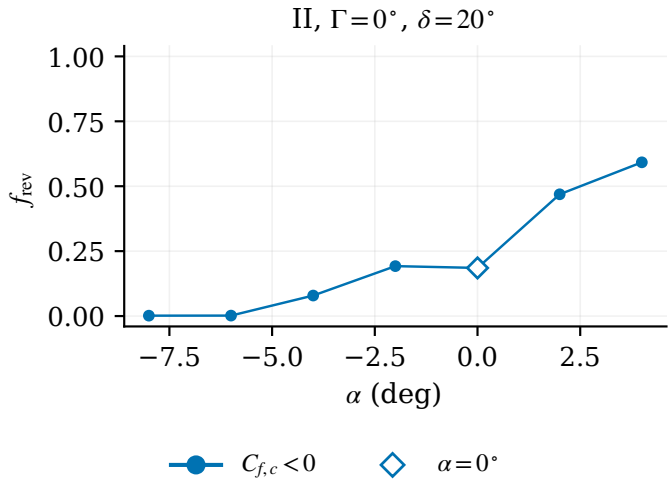



**Fig. 17 Forward-swept upper-control reversed-shear fraction at $\delta = 20^\circ$; the open diamond marks the $\alpha = 0^\circ$ solution used in the loading analysis.**

At $\alpha = 2^\circ$, II's reversed fraction is close to I's 49%, despite its much smaller control lift loss across $\alpha = 0 \rightarrow 2^\circ$. The differing pressure distributions and load locations explain why comparable reversed areas can accompany different lift and moment changes. Together, the two planform comparisons show how fixed-surface growth, control loss and their chordwise/spanwise locations combine in the integrated response.

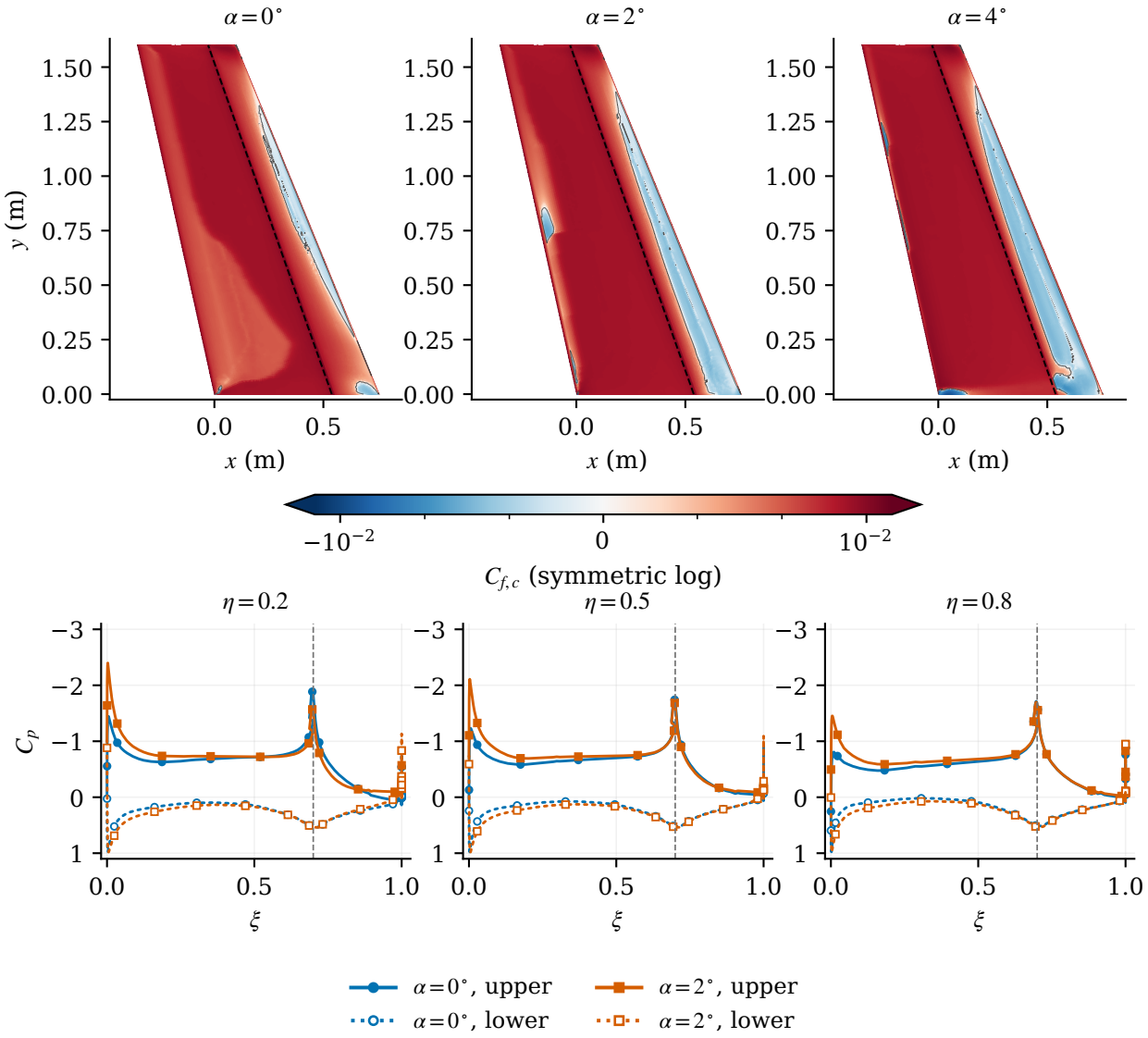



**Fig. 18 Forward-swept planform, $\delta = 20°$: upper-surface chordwise shear (top) and pressure sections across $\alpha = 0 \rightarrow 2°$ (bottom).**

## VI. Conclusions

The reference computation shows that continued total lift growth can conceal reduced growth on both the fixed surface and the full-span control. For configuration I at zero dihedral and $\delta = 20°$, fixed lift gain exceeds actual control lift loss, yet the fixed surface accounts for approximately 63% of the total departure below the preceding trend. The control contributes more strongly to the positive pitching-moment departure about the numerical aerodynamic center.

Pressure sections and regional integrations explain why stronger leading-edge suction is consistent with reduced fixed-surface growth: the forward-region gain falls below its preceding trend, while the intermediate region actually loses lift. Similar aft lift losses produce larger positive moment increments outboard as their downstream moment arms

grow. The upper-control shear pattern changes from a hinge-adjacent reversed band to trailing-edge-region reversal across the computed interval. Its subsequent upstream extension coexists with renewed total lift growth. Reversed-shear extent consequently complements the pressure and load evidence without determining the integrated response by itself.

The measurements establish distinct lift and moment responses across deflection and geometry. RANS reproduces the principal reference directions with a displaced lift interval and substantial coefficient-level bias, while correspondence is weaker for the dihedral and forward-swept cases. The detailed loading findings therefore characterize the computed states.

For matched dihedral, a large part of the absolute lift reduction is the reduced vertical component of panel-normal force, but adjacent lift increments reverse their ordering between the two orientations. The common-axis moment comparison also changes from reinforcement to compensation between the contributions from $F_x$ and $F_z$ across incidence intervals. Both the force magnitudes and their moment arms enter this incidence-dependent response.

The forward-swept planform gives stronger fixed-surface lift growth, much weaker control loss, positive intermediate-chord growth and an inboard-concentrated aft lift loss over the matched $\alpha = 0 \rightarrow 2^\circ$ interval. Its aft moment increments consequently lack the reference model's increasing positive outboard progression about the corresponding aerodynamic centers. This connects the different pitching responses to the chordwise and spanwise organization of the computed loading.

Future computations on the finer levels of the established mesh hierarchy are planned to assess the resolution sensitivity of the pre-stall load redistribution, with particular attention to pitching moment and the fixed- and control-surface contributions. The present joint analysis identifies how continued lift growth can coexist with different surface deficits and pitching-moment weights.

## A. Regional and Solution Checks

Figure 19 shows the two partitions used to check the regional interpretation. The first divides the semispan into thirds and the chord at the hinge; the second uses spanwise quarters and adds a chord boundary at $\xi = 0.4$.

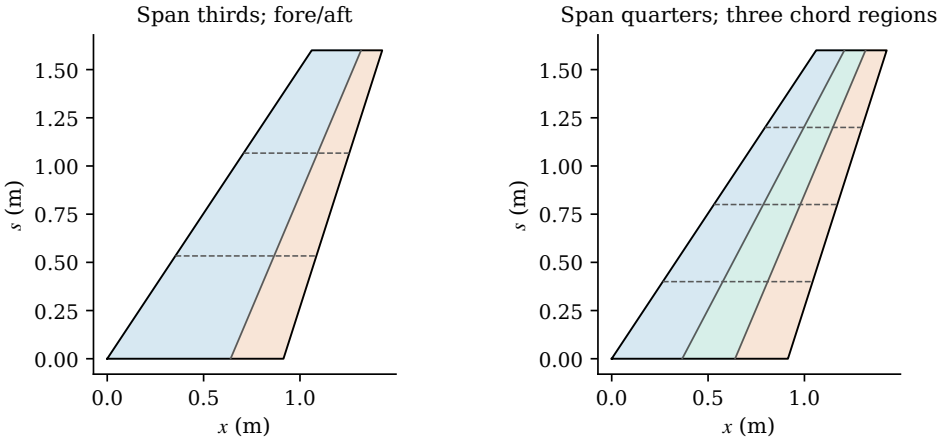



**Fig. 19 Alternative integration partitions: spanwise thirds with fore/aft regions (left), and quarters with forward/intermediate/aft regions (right).**

Moving the spanwise boundary from one third to one quarter changes the inboard aft departure from approximately −0.0127 to −0.00939, while complete fore/aft sums remain unchanged. Finer chord subdivisions preserve the two negative fixed-chord departures. These checks support the complete loading interpretation while showing that individual bin magnitudes depend on the partition.

Small-threshold and geometric-partition checks are retained in the reproducibility record; the main figures use $C_{f,c} < 0$.

At $\Gamma = 45°$, the stored upper-control reversed-shear fractions decrease between $\alpha = 2°$ and $\alpha = 4°$, from about 0.58 to 0.09 at $\delta = 20°$ and from 0.91 to 0.20 at $\delta = 30°$. Their physical origin is unresolved, so they are not interpreted as a transition or reattachment sequence. The integrated $\delta = 20°$ lift irregularity over $\alpha = -6 \rightarrow -4°$ exceeds the retained terminal variations of order $10^{-4}$, but this does not establish a unique steady state.

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## References

[1] Sears, R. I., and Liddell, R. B., "Wind-Tunnel Investigation of Control-Surface Characteristics VI: A 30-Percent-Chord Plain Flap on the NACA 0015 Airfoil," Tech. Rep. NACA-WR-L-454, National Advisory Committee for Aeronautics, 1942. URL https://ntrs.nasa.gov/citations/19930092872.

[2] Campbell, I. J., Blanks, C. F., and Leaver, D. A., "Aerodynamic Characteristics of Rectangular Wings of Small Aspect Ratio,"

Tech. Rep. Reports and Memoranda No. 3142, Aeronautical Research Council, 1956. URL https://reports.aerade.cranfield.ac.uk/handle/1826.2/3709.

[3] Harper, C. W., and Maki, R. L., "A Review of the Stall Characteristics of Swept Wings," Tech. Rep. NASA TN D-2373, National Aeronautics and Space Administration, 1964. URL https://ntrs.nasa.gov/citations/19640014908.

[4] Sivells, J. C., and Neely, R. H., "Method for Calculating Wing Characteristics by Lifting-Line Theory Using Nonlinear Section Lift Data," Tech. Rep. NACA Report 865, National Advisory Committee for Aeronautics, 1947. URL https://ntrs.nasa.gov/citations/19930091938.

[5] Goitia, H., and Llamas, R., "Nonlinear Vortex Lattice Method for Stall Prediction," *MATEC Web of Conferences*, Vol. 304, 2019, p. 02006. https://doi.org/10.1051/matecconf/201930402006.

[6] Rausa, A., Gori, G., Zanotti, A., Gibertini, G., Guardone, A., and Auteri, F., "Multifidelity Numerical–Experimental Database for Tailplane Aerodynamics," *AIAA Journal*, 2026, pp. 1–12. https://doi.org/10.2514/1.J066373, article in Advance.

[7] Palacios, F., Alonso, J., Duraisamy, K., Colonno, M., Hicken, J., Aranake, A., Campos, A., Copeland, S., Economon, T., Lonkar, A., Lukaczyk, T., and Taylor, T., "Stanford University Unstructured SU2: An open-source integrated computational environment for multi-physics simulation and design," Tech. Rep. AIAA Paper 2013-287, American Institute of Aeronautics and Astronautics, 2013. https://doi.org/10.2514/6.2013-287.

[8] Spalart, P. R., and Allmaras, S. R., "A One-Equation Turbulence Model for Aerodynamic Flows," Tech. Rep. AIAA Paper 1992-439, American Institute of Aeronautics and Astronautics, 1992. https://doi.org/10.2514/6.1992-439.

[9] Cakmakcioglu, S. C., Bas, O., Mura, R., and Kaynak, U., "A Revised One-Equation Transitional Model for External Aerodynamics," Tech. Rep. AIAA Paper 2020-2706, American Institute of Aeronautics and Astronautics, 2020. https://doi.org/10.2514/6.2020-2706.

[10] Van Leer, B., "Towards the ultimate conservative difference scheme. V. A second-order sequel to Godunov's method," *Journal of computational Physics*, Vol. 32, No. 1, 1979, pp. 101–136.

[11] Venkatakrishnan, V., "On the accuracy of limiters and convergence to steady state solutions," Tech. Rep. AIAA Paper 1993-880, American Institute of Aeronautics and Astronautics, 1993. https://doi.org/10.2514/6.1993-880.